Video coding standards, MPEG 1, 2, H.264/AVC, H.265/HEVC, H.266/VVC, VP8, VP9, AV1, Dirac, ProRes, ...

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Contents

- **Introduction**
  - Development of **major video coding standards**
- **MPEG video compression standards**
  - **Hybrid DPCM/DCT encoder** in **MPEG 1** and **MPEG 2 standards**
    - **Interframe prediction** and **motion compensation**
    - **Basic features** of MPEG 1, MPEG 2 and MPEG 4
  - **H.264**/Advanced Video Coding (**AVC**)
    - **Multiview video coding** (**MVC**) and **UHDTV**
  - **H.265**/High Efficiency Video Coding (**HEVC**)
  - **H.266**/Versatile Video Coding (**VVC**)
- **Other video compression formats**
  - **VP8** and **VP9**, **AOMedia Video 1** (**AV1**), **Apple ProRes**, **Dirac** and **Dirac Pro** (**VC2**), ...

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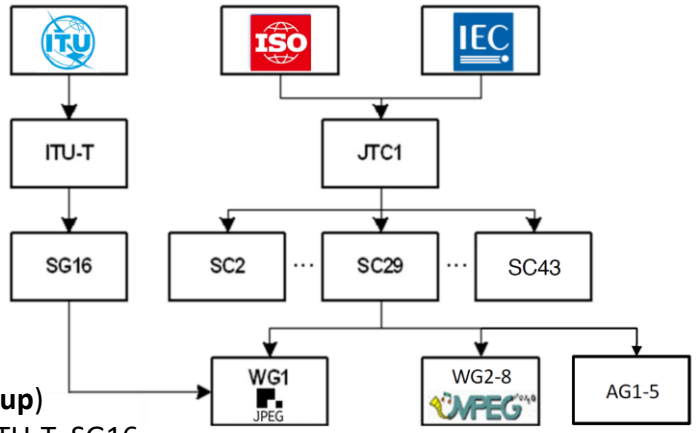
Introduction

Standardization in **image** and **video coding**

- Major **standardization committees** in **this field**
  - **Joint Photographic Experts Group (JPEG)**
  - **Moving Picture Experts Group (MPEG)**



JPEG (Joint Photographic Experts Group)
ISO/IEC JTC 1/SC 29/**WG 1** and ITU-T SG16

MPEG (Moving Picture Experts Group)
ISO/IEC JTC 1/SC 29/**WG 2-8** (WG11) and ITU-T SG16

- **Both groups awarded multiple times** (e.g. Emmy, ...)
  - **Television Academy** (Technology and Engineering Emmy Awards)

Emmy Award
MPEG and JPEG
(1995-96)

Emmy Award
JPEG „JPEG 1992“
(2019)

71st Engineering Emmy Award
for **JPEG 1**
(23. 10. 2019
Los Angeles, USA)

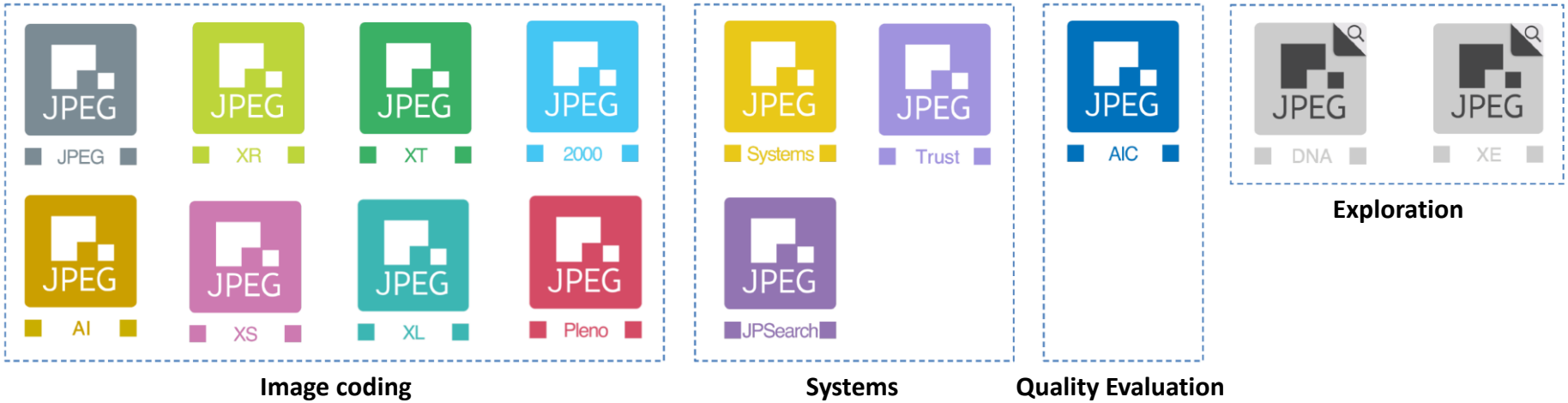
77th Engineering Emmy Award
for **JPEG XS**
(14. 10. 2025
Los Angeles, USA)

Introduction

JPEG family of image coding standards

JPEG (Joint Photographic Experts Group)

- Since **1992** and **„JPEG 1“ several** other **major image coding standards introduced** by **JPEG**



- **JPEG** „JPEG 1, Legacy JPEG“ **1992** (ISO/IEC 10918)
  - One of **most widespread standards** ever **„human heritage standard“** (**~3.3 billion images** shared daily via social networks, **mostly JPEG files**)
  - Exploiting **block-based transform coding DCT**
- **JPEG 2000** (ISO/IEC 15444)
  - Exploiting **DWT**, applications in **specific areas** (**DCI**, **DICOM**, **PDF**, ...)

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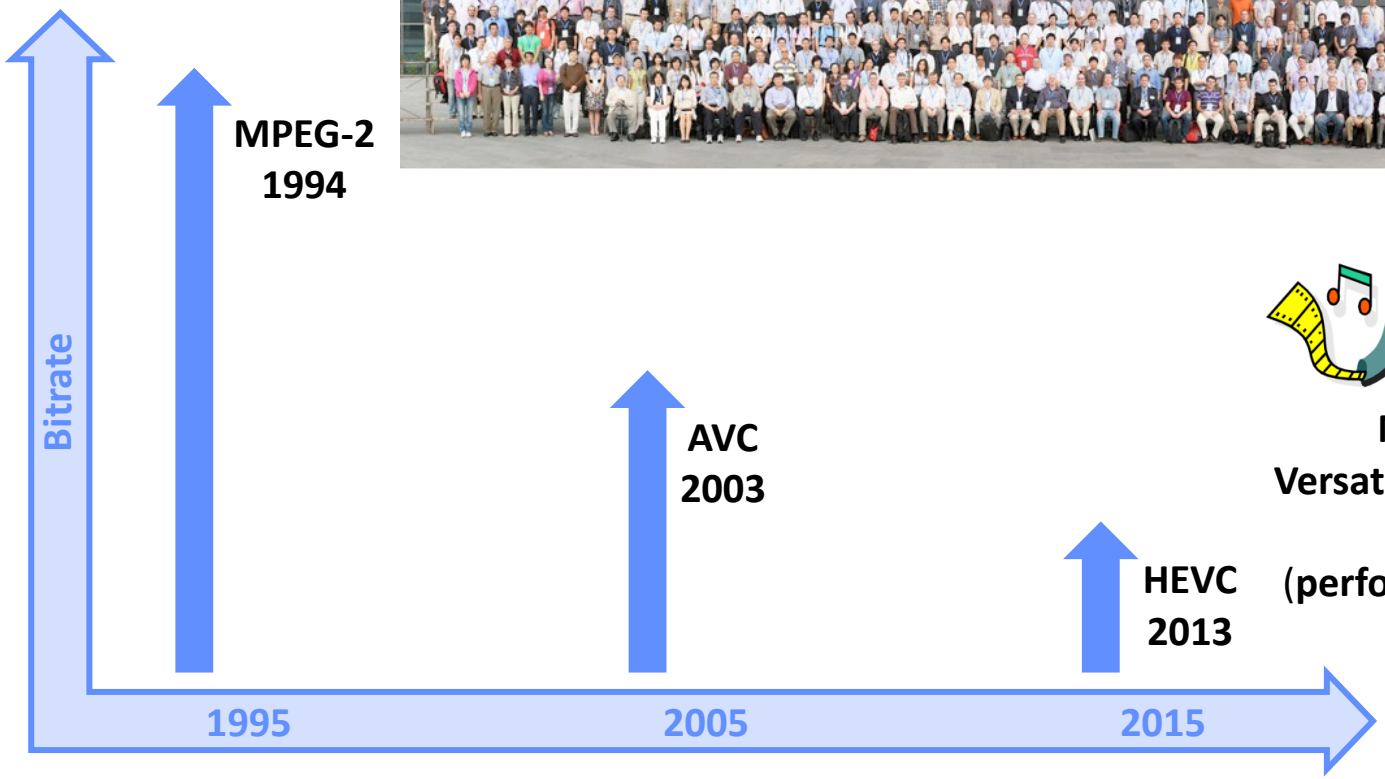
Introduction

Video coding performance

- Every **10 years** new standard and **50% bitrate savings**



Newest standard
Versatile Video Coding (VVC)
July 2020
(performance gain 30-50 %)

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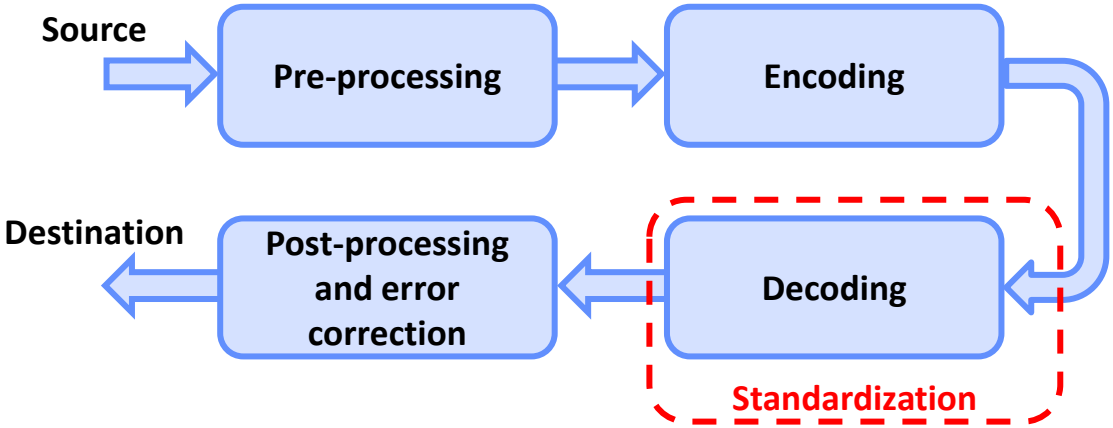
Introduction

- Approximately **10 years development cycle**
  - **ITU-T** Video Coding Experts Group (ITU-T – **VCEG**)
    - **H.261** (1990), **H.263** (1995) → **videoconferencing, low bitrates**
  - **ISO** Moving Picture Experts Group (ISO/IEC – **MPEG**)
    - **MPEG-1** (1993), **MPEG-2** (1996) → **DVD**, **DVB**
  - **ITU-T/ISO** Joint Video Team (**JVT**)
    - **H.264/MPEG-4 Part 10 AVC** (Advanced Video Coding)
      - First version **2003**, **twice more efficient** than MPEG-2 → DVB and **HDTV**
  - **ITU-T/ISO** Joint Collaborative Team on Video Coding (**JCT-VC**)
    - **H.265/MPEG-H Part 2 HEVC** (High Efficiency Video Coding)
      - First version **2013**, **twice more efficient** than AVC → **UHDTV applications**
  - **ITU-T/ISO** Joint Video Experts Team (**JVET**)
    - **H.266/VVC** (Versatile Video Coding)
      - First version **July 2020**, 30-50% **better efficiency** than HEVC

Introduction

- **Bitstream**, **Syntax**, and **Decoder** are **standardized**
  - **Encoder optimization**
    - Allows **encoder optimization** while keeping **compatibility**
  - **Implementability**
    - Allows **encoder simplification** (lower coding efficiency)
  - **Disadvantage**
    - Provides **no guarantees** of **quality**



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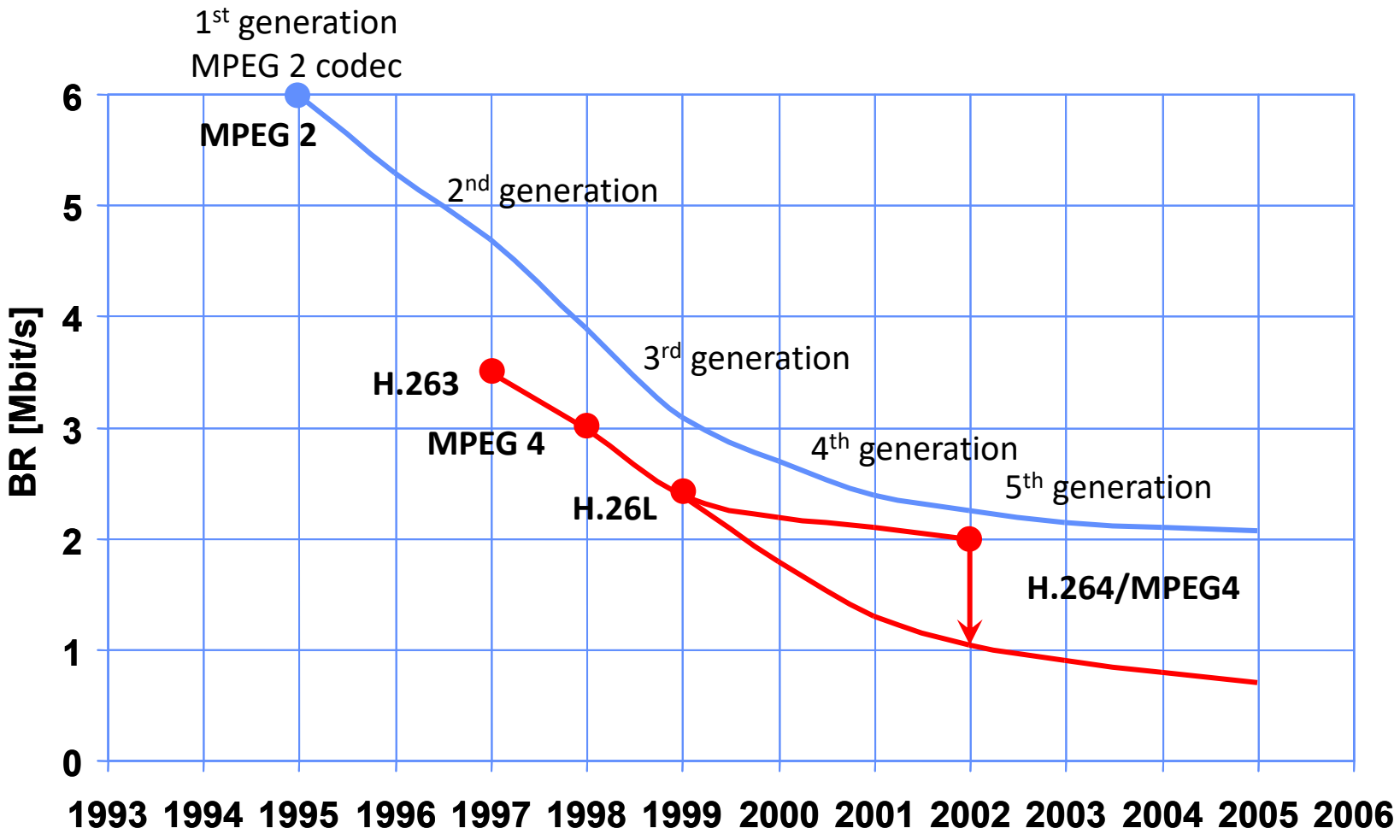
Introduction

- **Bit-rate development** for **standard resolution video**
  - For SD video 1995-2005



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Introduction

Approximate formula for bitrate estimation

$$B \approx A \cdot V \quad [\text{Mbps}]$$

- Coding **technology**
  - **A = 4** **MPEG-2** (1994)
  - **A = 2** **AVC** (2003)
  - **A = 1** **HEVC** (2013)
  - **A = ½** **VVC** (2020)
- **Video format** (resolution)
  - **V = 1** **SDTV** (720×576, 25i nebo 720×480, 30i)
  - **V = 4** **HDTV** (1920×1080, 25i)
  - **V = 16** **UHDTV** (3840×2160, 50p)
- **Examples**
  - **SD** (570/480, 25 fps) and **MPEG-2** → **4 Mbps**
  - **HD** (1080, 25 fps) and **AVC** → **8 Mbps**
  - **UHD** (2160, 50 fps) and **HEVC** → **16 Mbps**
  - **UHD** (2160, 50 fps) and **VVC** → **8 Mbps**

Domański, M., et al. Approximate video bitrate estimation for television service, Doc. MPEG M3671, Warsaw, 2015.

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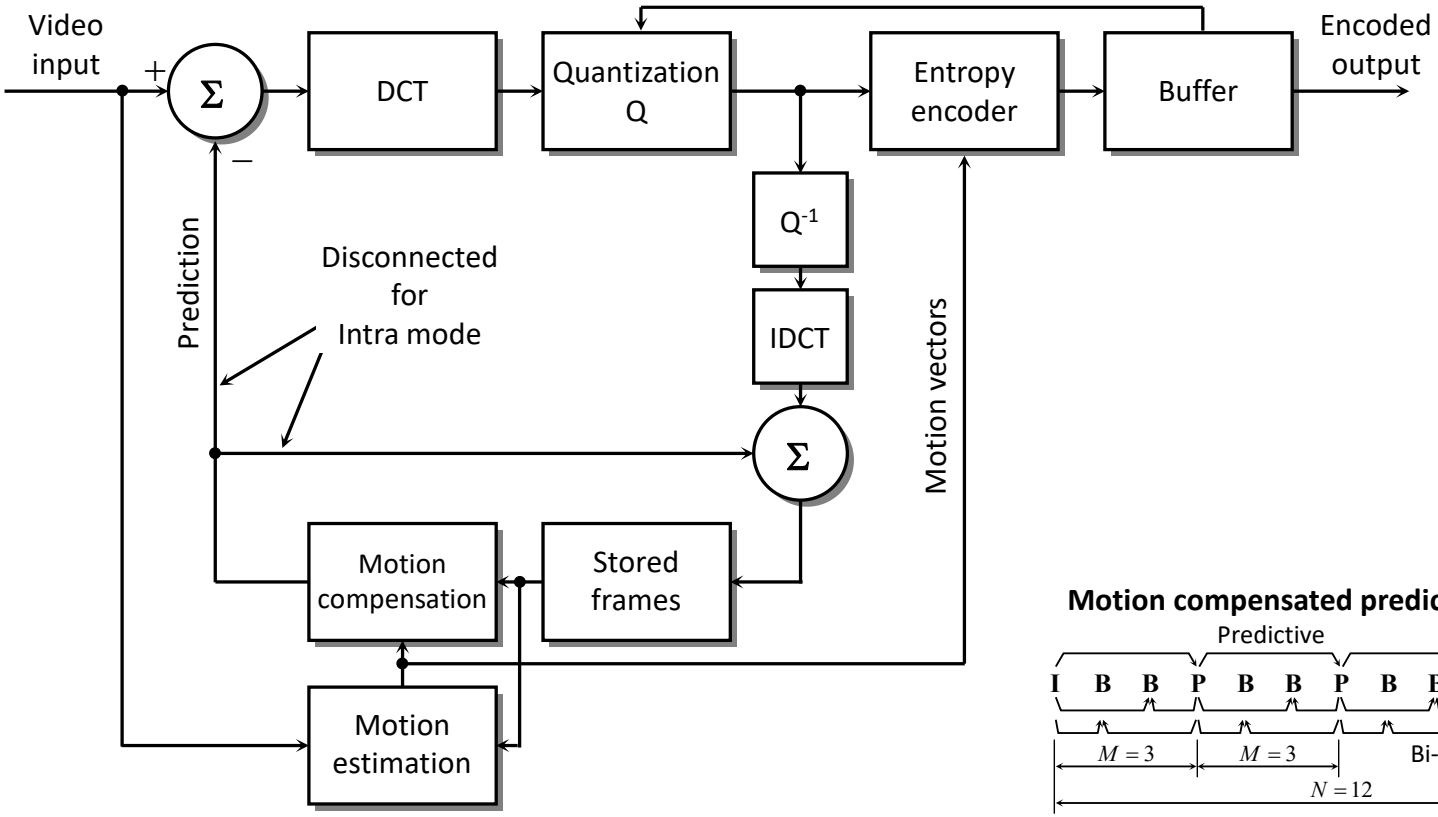
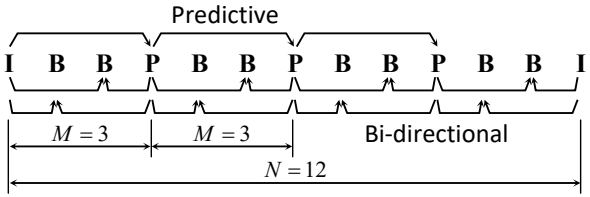
MPEG 1, 2, …

Utilization in hybrid DPCM/DCT encoder



Motion compensated prediction in MPEG 2



- **Identical general model** for **majority** of **encoders**
  - **Motion compensated prediction** (DPCM), **transform coding**, **quantization** and **entropy encoder**

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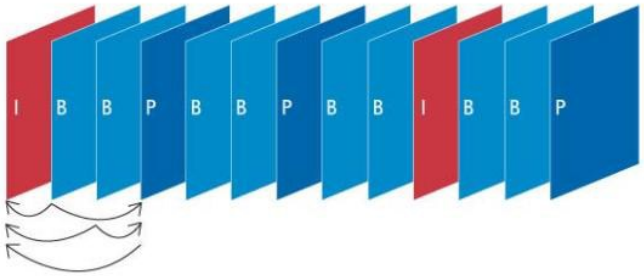
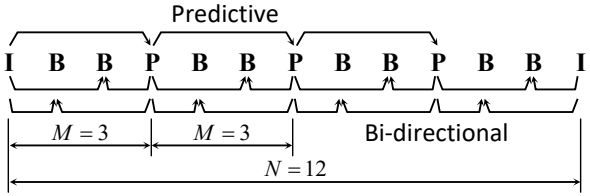
MPEG 1, 2, …

Motion compensated prediction (MPEG 1, 2, …)

- **Video frame sequence** – frames **not encoded separately**
- Exploiting **different frame types**
  - **I** (**Intra-frame**) – **complete image** does not depend on neighborhood, compressed using „**JPEG compression**“
  - **P** (**Predictive**) – residual, **predicted frame** based on **last I** or **P frame**, can be **used** as **reference**
  - **B** (**Bi-directional**) – residual, „**bi-directionally**“ **predicted frame**, predicted from **two reference frames** (**past** and **future**)



Motion compensated prediction in MPEG 2



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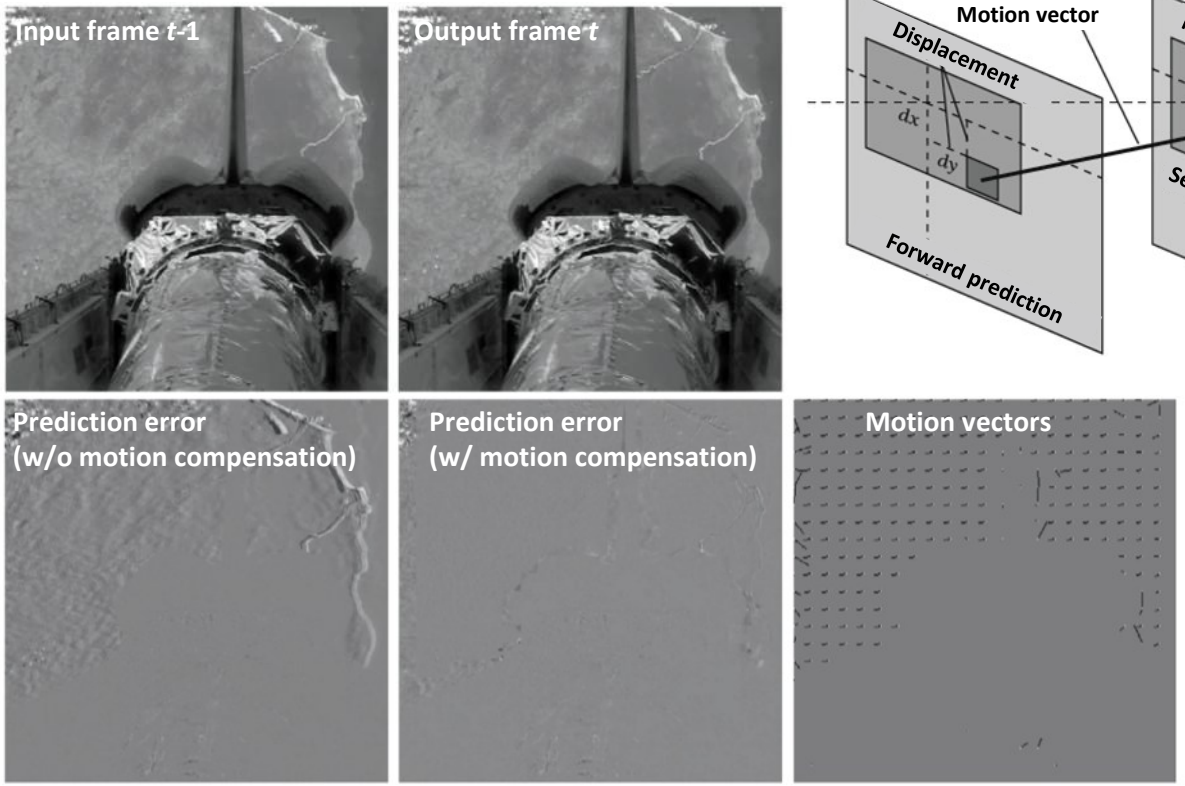
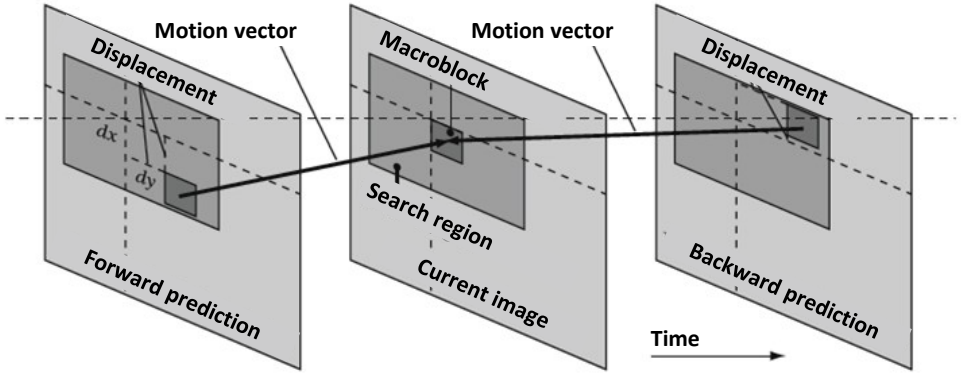
MPEG 1, 2, …

- **Motion compensated prediction (MPEG 1, 2, …)**
  - **Block based motion compensation** and **motion vectors**
    - **I** frames (**intraframe** coding)
    - **P**, **B** frames (**interframe** coding with **motion compensation**)





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MPEG 1, 2, ...

Motion compensated prediction (MPEG 1, 2, ...)

- **Decoding** video bitstream **frame** by **frame**
  - Decoding starts always with **I frame** – **independent** (intra coded)
  - **P/B frames** – **information** on **changes** in respect to reference frame
  - **B frames** – **frames** in bitstream **saved** in **different order**

- **GOP - Group of Pictures**
  - **Group** of **pictures** between two **I frames** (length typically 3 to 12)
  - Example:
    - GOP = 4, e.g., IPPP IPPP ...
    - GOP = 15, e.g., IPPPPPPPPPPPPPP IPPPPPPPPPPPPPP ...
    - GOP = 8, e.g., IBPBPBPB IBPBPBPB ...
  - **Lowering I frames frequency** – **reducing** (**encoded**) **video bitrate**
  - **Removing B frames** – **reducing latency**
    (**playback delay** due to **more complex decoding**)

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MPEG 1, 2, ...

- **MPEG 1** (ISO/IEC 11172, **1993**)
  - Part 1-5, **video coding Part 2** (**1993**)
  - **Developed** from **ITU-T Rec. H.261**, first **useful standard**
  - Coding **SD video**, **CD** (VCD) **media**, **progressive** video, **VHS quality**, bitrate **~1.5 Mbps**, ...
- **MPEG 2** (ISO/IEC 13818, **1996**)
  - Part 1-11, **video coding Part 2** (ITU-T Rec. **H.262**, **1996**)
  - **Better efficiency** for **better quality** and **higher bitrates**
  - Intended for **DVD**, **DVB**, **interlaced**, and **progressive** video, ...
- **MPEG 4** (ISO/IEC 14496, **1999**)
  - Part 1-33, **video coding Part 2 Visual** (**1999**)
  - **Part 10 Advanced Video Coding** (**2003**)
  - Intended for **HDTV** applications, **DVB**, **Blu-ray**, ...

H.264/Advanced Video Coding

- **Video coding** in **H.264/AVC** standard
  - Based on **hybrid video coding**
  - Basic features **similar** to **previous standards**
- New **key features H.264/AVC** in **respect** to **MPEG2**
  - Enhanced **motion compensation**
  - **Integer transform** of smaller blocks
  - Enhanced **intraframe prediction**
  - Adaptive **de-blocking filter**
  - Enhanced **entropy coding**
- **Substantial bitrate savings** relative to **other standards**
  - Maintaining **same image quality**
  - **No particular block** responsible for the **substantial savings** – **many partial improvements**

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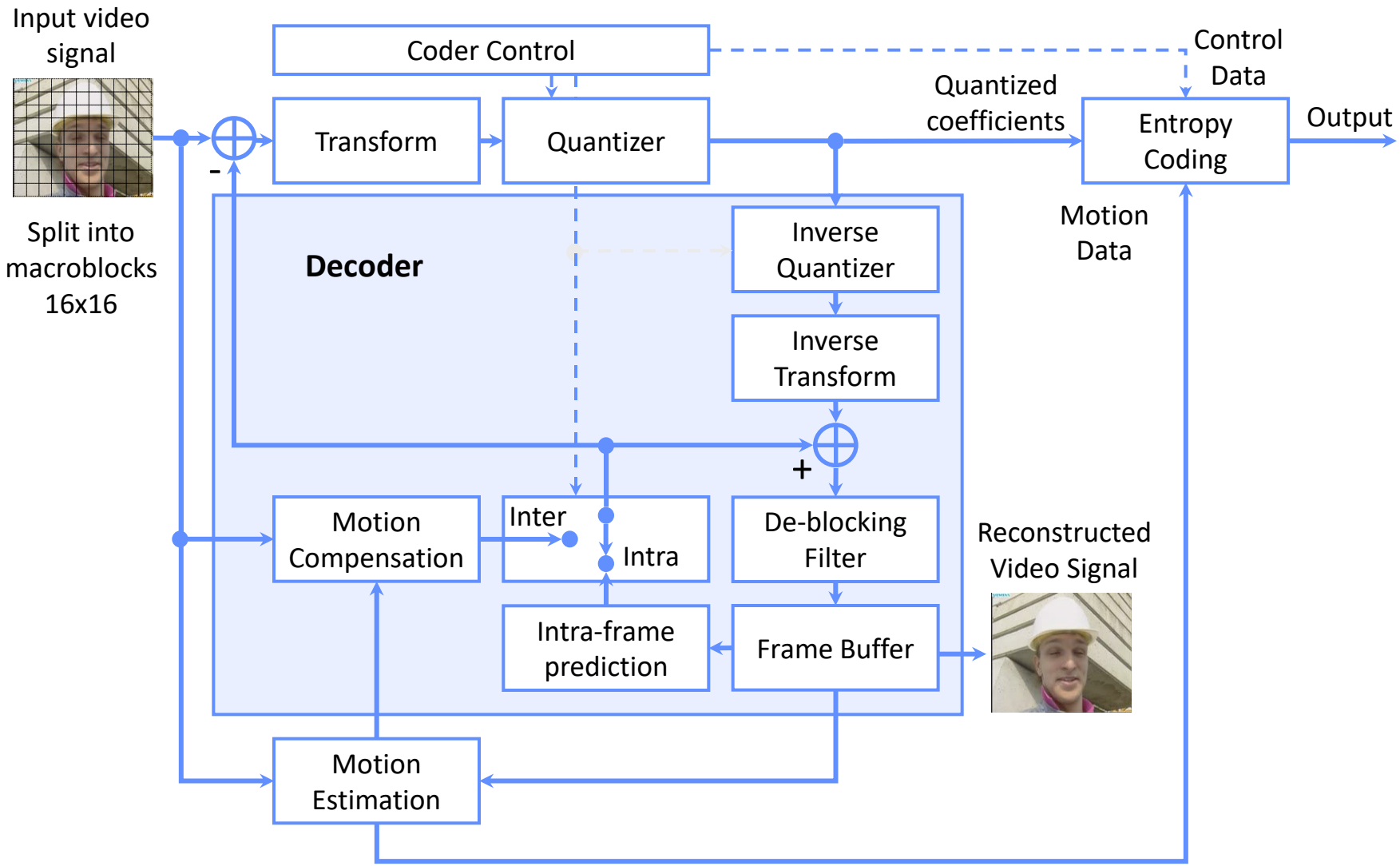
H.264/Advanced Video Coding



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H.264/Advanced Video Coding

- **Profiles** and **levels** define **different configurations**
  - **Profile** – **set** of **algorithmic features** to **create bitstream**
  - **Level** – **degree** of **capability/limits** (**resolution**, **bitrate**)
- **Standard H.264/AVC** defined originally **three profiles**
  - **Baseline** – lower capability/quality, high error resilience (e.g. videoconferencing, mobile video)
  - **Main** – high compression quality (e.g. broadcasting)
  - **eXtended** – features for efficient streaming
- **Standard H.264/AVC** defined originally **eleven levels**
  - Define upper limits of **frame resolution**, **processing rate**, etc.

H.264/Advanced Video Coding

Informative overview of levels and parameters

Level	Maximum decoding speed [macroblocks/s]	Maximum frame size [macroblocks]	Maximum video bitrate [kbits/s] (Constrained Baseline, Baseline, Extended, Main Profiles)	Examples for high resolution and highest frame rate (maximum stored frames)
1	1,485	99	64	176×144@15.0 (4)
1b	1,485	99	128	176×144@15.0 (4)
1.1	3,000	396	192	352×288@7.5 (2)
1.2	6,000	396	384	352×288@15.2 (6)
1.3	11,880	396	768	352×288@30.0 (6)
2	11,880	396	2,000	352×288@30.0 (6)
2.1	19,800	792	4,000	352×576@25.0 (6)
2.2	20,250	1,620	4,000	720×576@12.5 (5)
3	40,500	1,620	10,000	720×576@25.0 (5)
3.1	108,000	3,600	14,000	1,280×720@30.0 (5)
3.2	216,000	5,120	20,000	1,280×1,024@42.2 (4)
4	245,760	8,192	20,000	2,048×1,024@30.0 (4)
4.1	245,760	8,192	50,000	2,048×1,024@30.0 (4)
4.2	522,240	8,704	50,000	2,048×1,080@60.0 (4)
5	589,824	22,080	135,000	3,672×1,536@26.7 (5)
5.1	983,040	36,864	240,000	4,096×2,304@26.7 (5)
5.2	2,073,600	36,864	240,000	4,096×2,304@56.3 (5)
6	4,177,920	139,264	240,000	8,192×4,320@30.2 (5)
6.1	8,355,840	139,264	480,000	8,192×4,320@60.4 (5)
6.2	16,711,680	139,264	800,000	8,192×4,320@120.9 (5)

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H.264/Advanced Video Coding

Image Partitioning

Macroblock slices

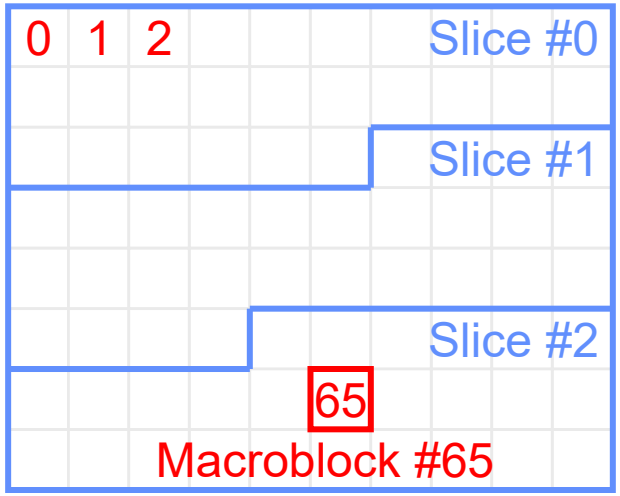
- Image split into **one** or **more slices**
- **Slices** are **independent**
- **Slices** consist of **macroblock sequences**

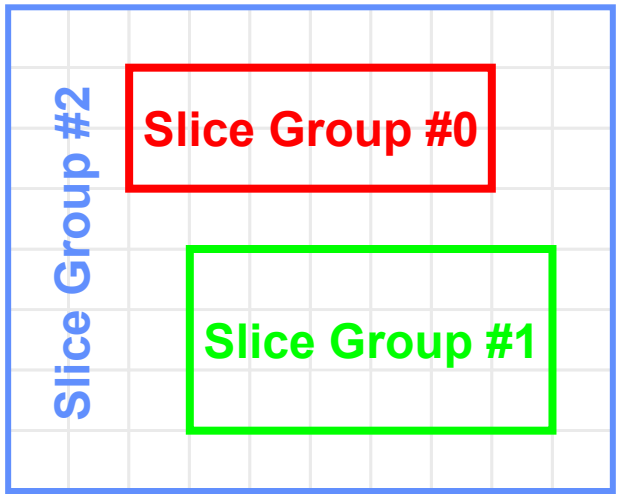
Macroblocks

- Contain **16x16 luma** and **2x (8x8) chroma** samples
- **Macroblocks** within slice **dependent**
- Possible further partitioning

Slice group

- **Pattern** of **macroblocks** defined based on **macroblock allocation maps**





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H.264/Advanced Video Coding

- **Intraframe prediction**
  - **Performed** on **available samples**
    - Previously **reconstructed** – **available** on **decoder side**
    - **Available** within **same slice**
- **Intraframe prediction** for **luma samples**
  - **16 individual predictions** for **4x4 blocks**
    - **9 modes** (8 directional and DC)
  - **1 prediction** for **whole macroblock 16x16**
    - **4 modes** (vertical, horizontal, DC and plane)
- **Intraframe prediction** for **chroma samples**
  - **Single prediction mode** for **both 8x8 chroma partitions**
    - **4 modes** (vertical, horizontal, DC and plane)

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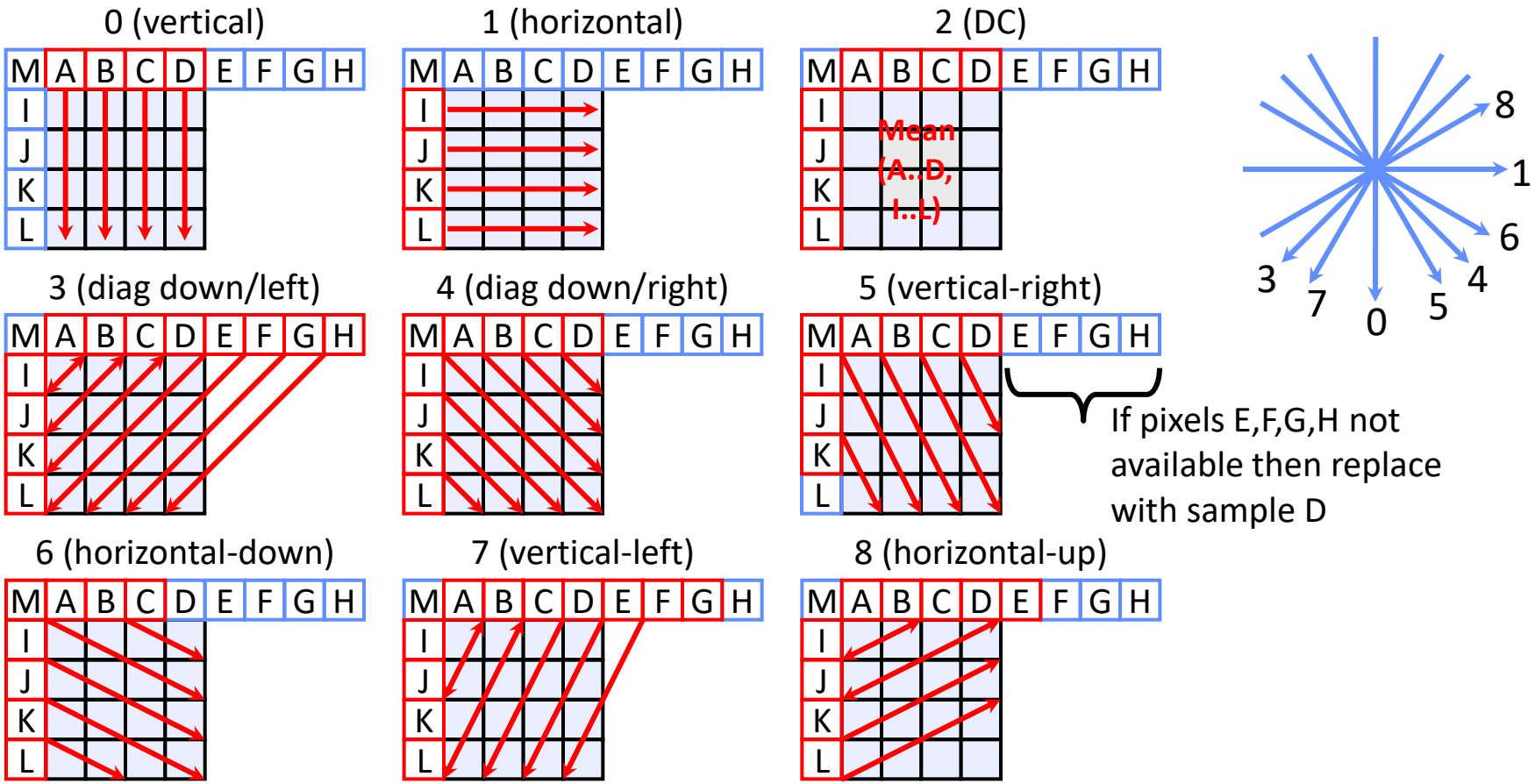
H.264/Advanced Video Coding

- **Intraframe luma prediction** – 4x4 blocks (main idea)
  - In total **9 modes** (**DC** + 8 **directional**)



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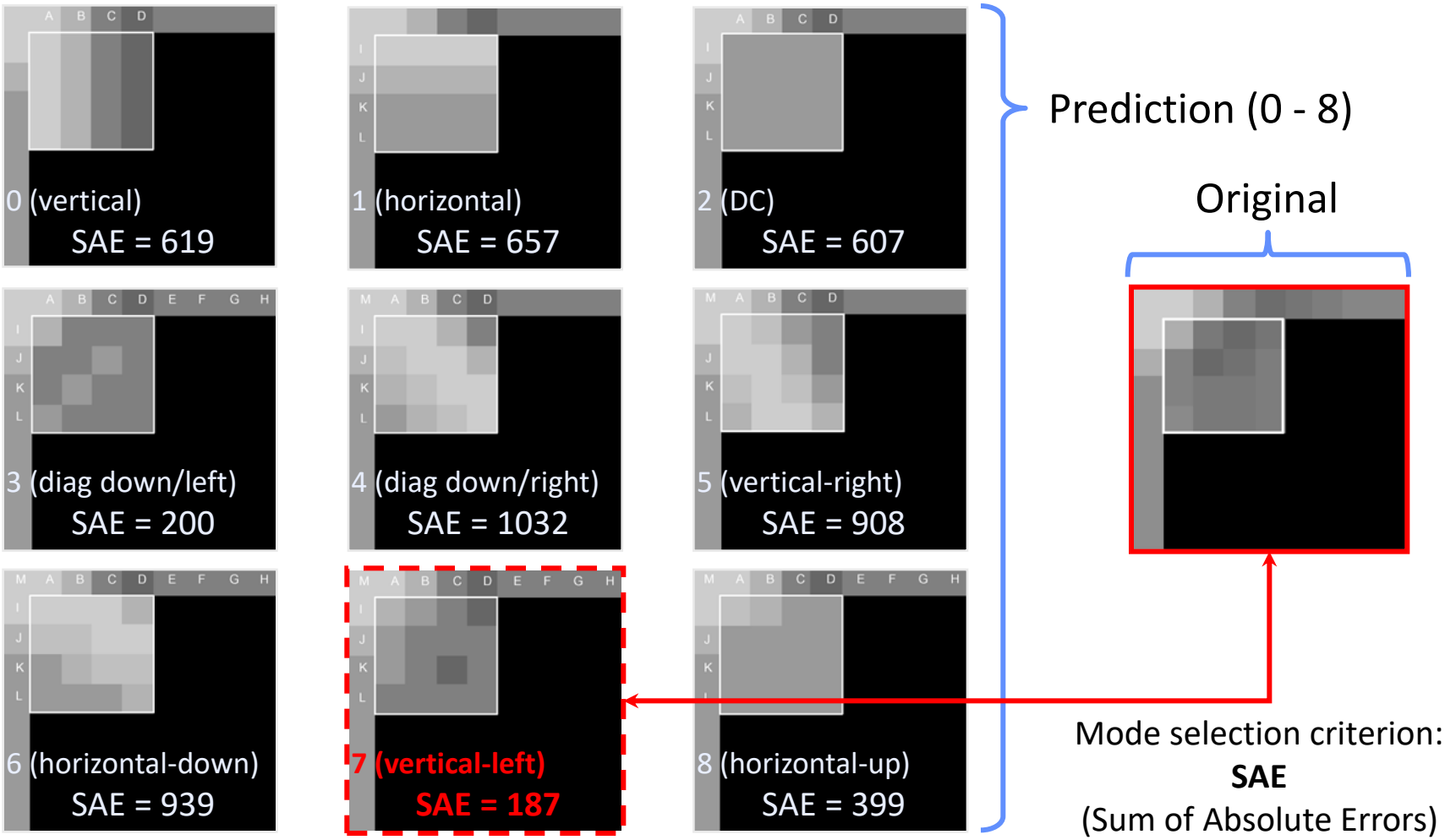
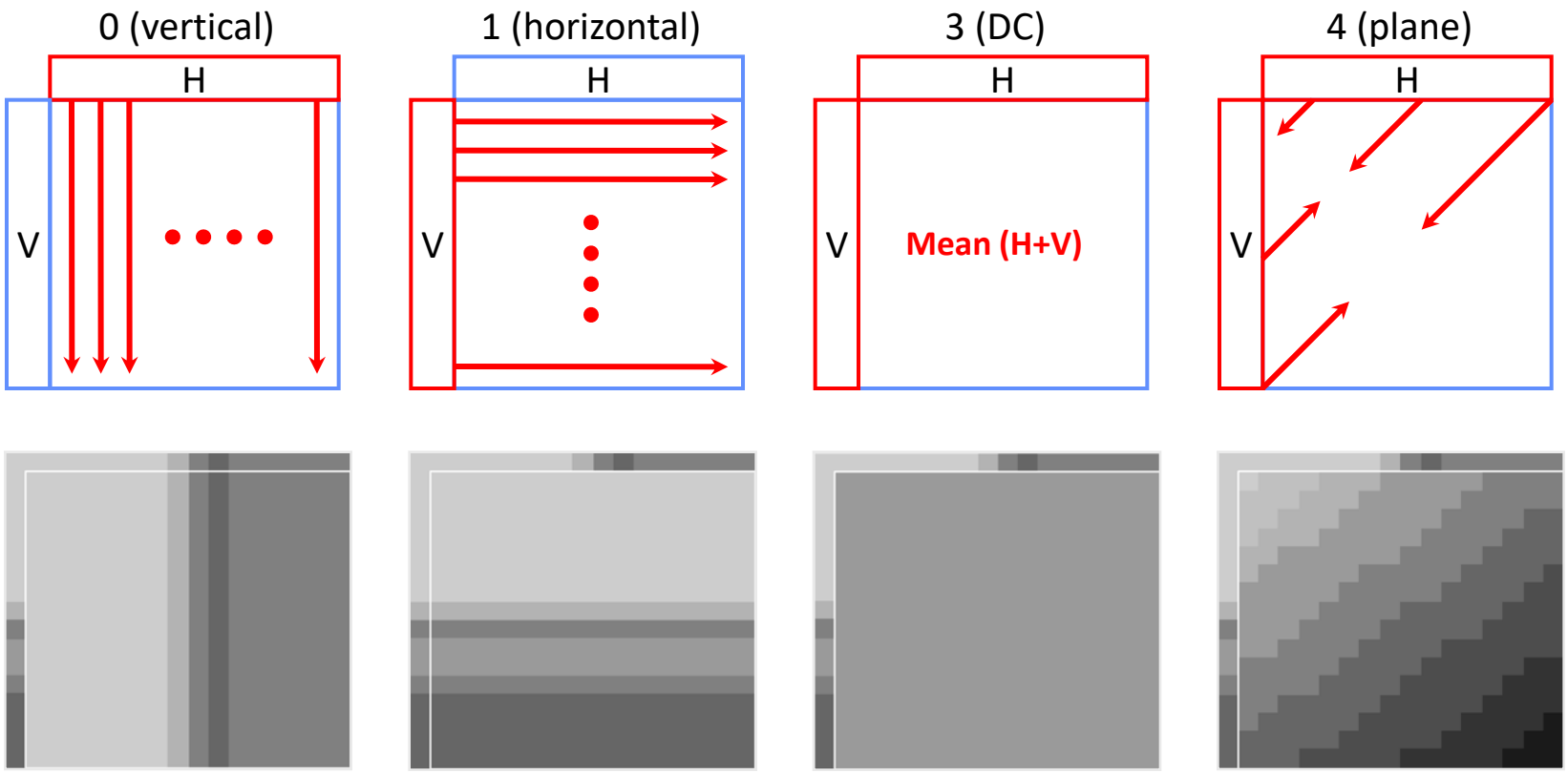
H.264/Advanced Video Coding

Intraframe luma prediction – 4x4 blocks (main idea)



H.264/Advanced Video Coding

- **Intraframe luma prediction** – 16x16 macroblocks
  - In total **4 modes** (vertical, horizontal, DC, plane)



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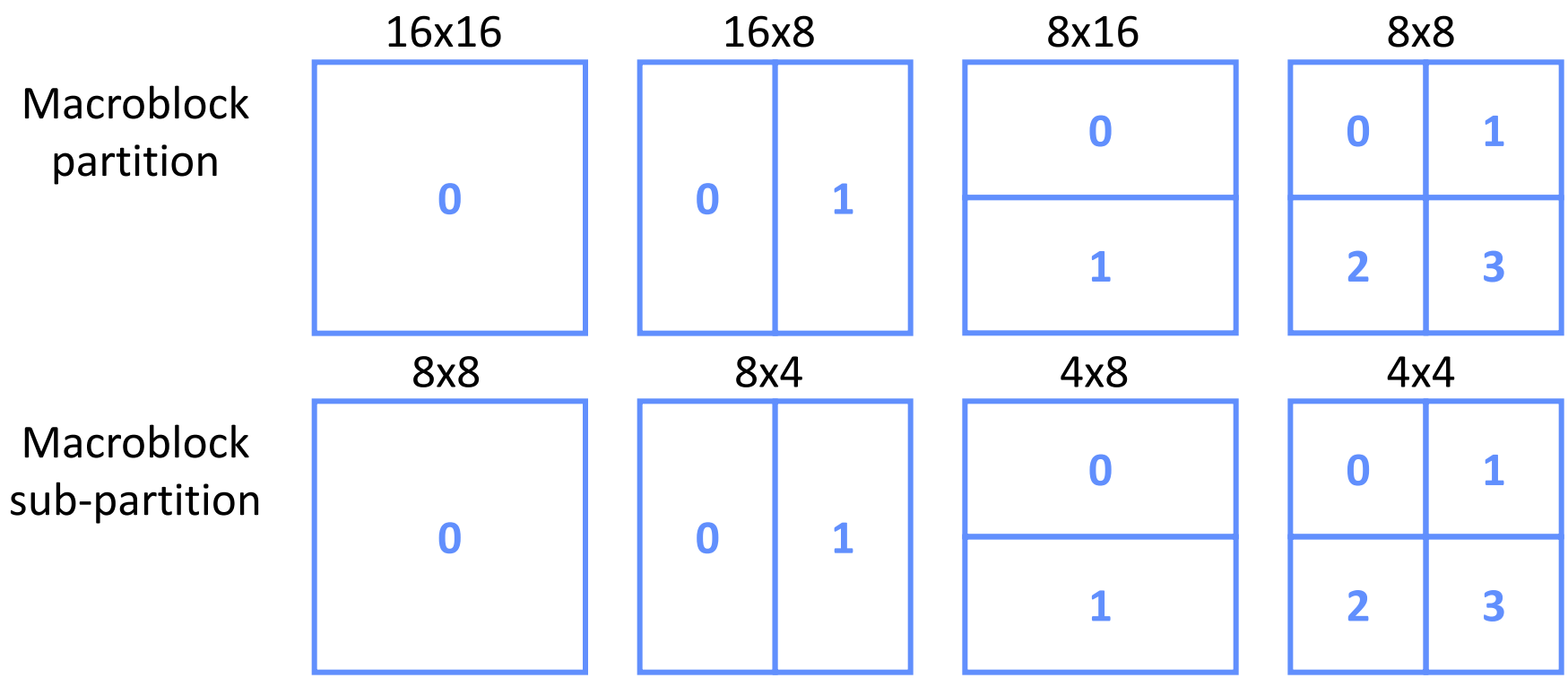
H.264/Advanced Video Coding

- **Interframe prediction** with **motion compensation**
  - **Variable luma block size** with **motion compensation** form **16x16** to **4x4** + variety of combinations



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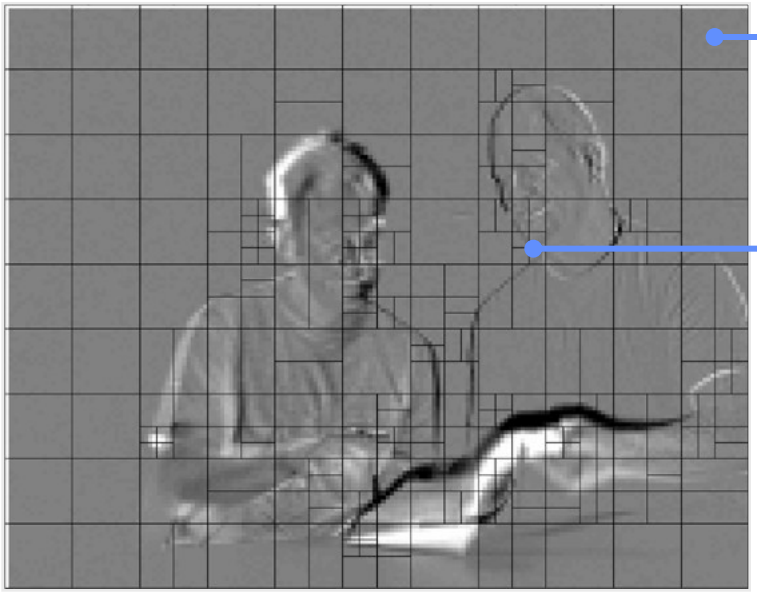
H.264/Advanced Video Coding

- **Interframe prediction** with **motion compensation**
  - **Motion vector needed** for **each partition** and **sub-partition**
  - **Coding efficiency highly dependent** on **partition size**
    - **Small** partition vs. **Large** partition – **Small residual error** vs. **Small number of motion vectors**
    - Motion vectors – ¼ sample accuracy

Residual frame



Small residual error:
- Large partition (16x16)

Large residual error:
- Small partition (4x4)

Codec chooses
suitable partition size

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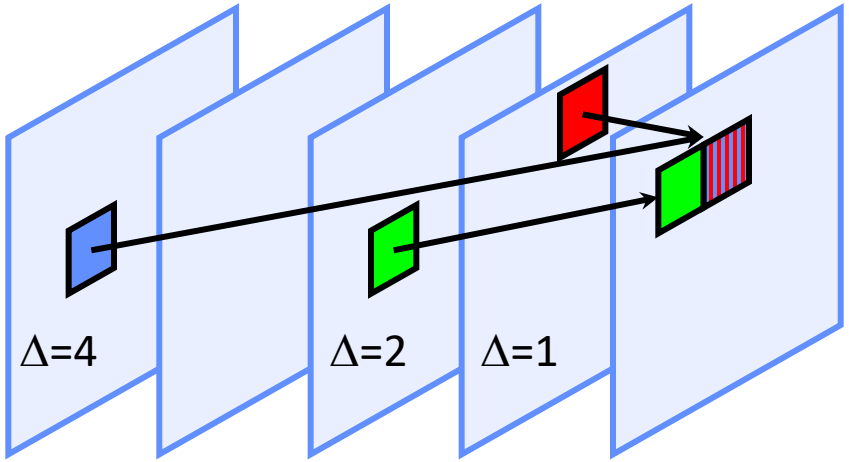
H.264/Advanced Video Coding

- **Interframe prediction** with **motion compensation**
  - **Compensated prediction** with **multiple reference frames**
    - **Better** motion **compensation efficiency**
    - More **complex encoder** and **decoder** (multiple frame buffer needed)



4 reference frames – already decoded frames

Current frame

Each block has its own **reference frame**

Reference frames must be **available** at **decoder**

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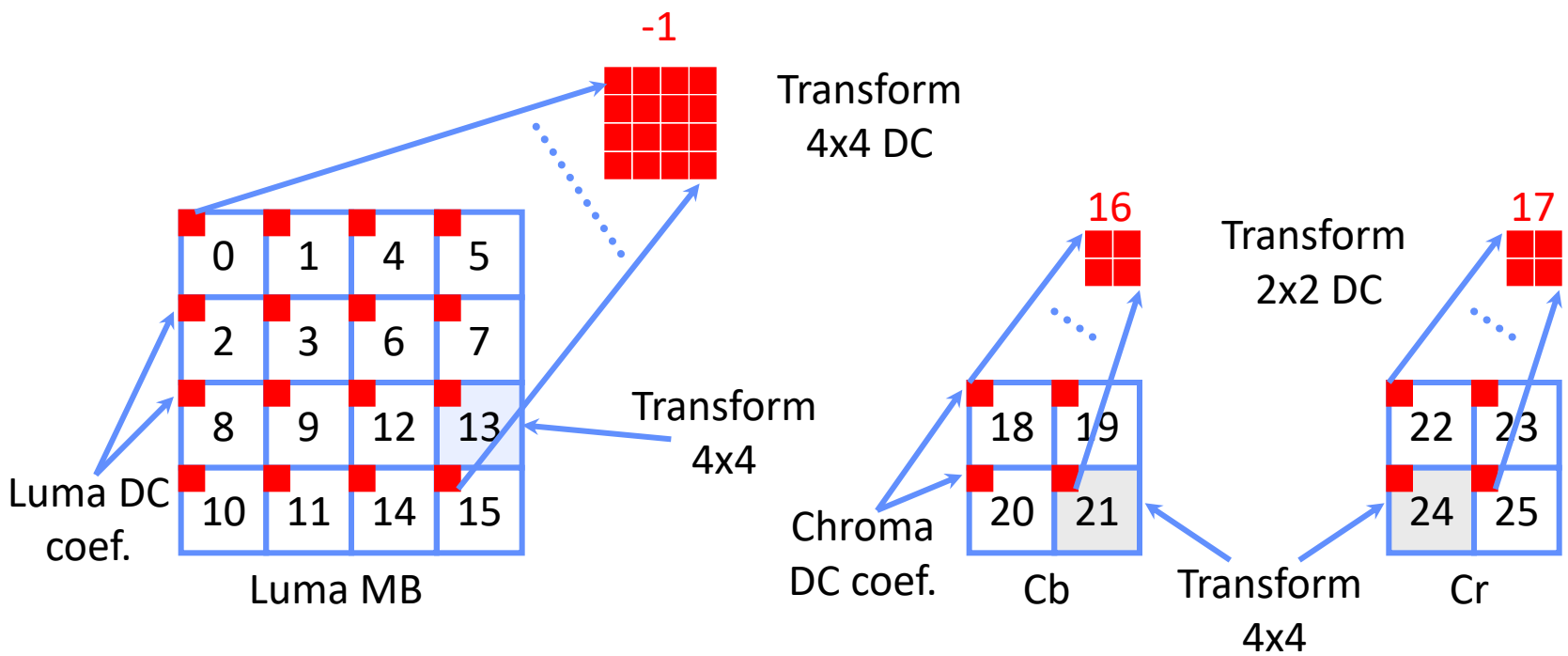
H.264/Advanced Video Coding

Transformation for residual error encoding

- Three **transform applications** (blocks **8x8** or **4x4**)
  - (**A**) Residual **blocks 4x4** (**integer transform** based on **DCT**)
  - (**B**) **4x4 luma** DC coefficients (**Hadamard** transform)
  - (**C**) **2x2 chroma** DC coefficients (**Hadamard** transform)
- Example **transform application** and **scanning order**



H.264/Advanced Video Coding

- **(A) Residual block transform** (example for 4x4)
  - **Integer transform** based on **DCT**
    - **Implementation** using **shifts** and **additions** (fast calculations)
    - **Multiplication** (**scaling**) embedded into **quantizer**
      - **Advantage** – **without rounding errors**
      - **Disadvantage** – **lower** compacting **efficiency** than **DCT**

- **Integer transform** derivation from **4x4 DCT**
  - **(1)** 4x4 DCT input matrix **X**

$$\mathbf{Y} = \mathbf{A}\mathbf{X}\mathbf{A}^T = \begin{bmatrix} a & a & a & a \\ b & c & -c & -b \\ a & -a & -a & a \\ c & -b & b & -c \end{bmatrix} [\mathbf{X}] \begin{bmatrix} a & b & a & c \\ a & c & -a & -b \\ a & -c & -a & b \\ a & -b & a & -c \end{bmatrix}$$

$$a = \frac{1}{2}$$

$$b = \sqrt{\frac{1}{2}} \cos\frac{\pi}{8}$$

$$c = \sqrt{\frac{1}{2}} \cos\frac{3\pi}{8}$$

H.264/Advanced Video Coding

- **(2)** 4x4 DCT **factorization**

$$\mathbf{Y} = \left(\mathbf{C}\mathbf{X}\mathbf{C}^T\right) \otimes \mathbf{E} \qquad a = \frac{1}{2}; b = \sqrt{\frac{1}{2}}\cos\frac{\pi}{8}; d = \frac{c}{b} = 0,414\ldots$$

$$= \left(\begin{bmatrix} 1 & 1 & 1 & d \\ 1 & d & -1 & -1 \\ 1 & -d & -1 & 1 \\ 1 & -1 & 1 & -d \end{bmatrix} [\mathbf{X}] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & d & -d & -1 \\ 1 & -1 & -1 & 1 \\ d & -1 & 1 & -d \end{bmatrix}\right) \otimes \begin{bmatrix} a^2 & ab & a^2 & ab \\ ab & b^2 & ab & b^2 \\ a^2 & ab & a^2 & ab \\ ab & b^2 & ab & b^2 \end{bmatrix}$$

- **(3) Core transform** modification (**CXC**^{T}) – **integer arithmetic**

$$\mathbf{Y} = \left(\mathbf{C}_f\mathbf{X}\mathbf{C}_f^T\right) \otimes \mathbf{E}_f \qquad a = \frac{1}{2}; b = \sqrt{\frac{2}{5}}; d = \frac{1}{2}$$

$$= \left(\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -1 & -2 \\ 1 & -1 & -1 & 1 \\ 1 & -2 & 2 & -1 \end{bmatrix} [\mathbf{X}] \begin{bmatrix} 1 & 2 & 1 & 1 \\ 1 & 1 & -1 & -2 \\ 1 & -1 & -1 & 2 \\ 1 & -2 & 1 & -1 \end{bmatrix}\right) \otimes \begin{bmatrix} a^2 & ab/2 & a^2 & ab/2 \\ ab/2 & b^2/4 & ab/2 & b^2/4 \\ a^2 & ab/2 & a^2 & ab/2 \\ ab/2 & b^2/4 & ab/2 & b^2/4 \end{bmatrix}$$

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H.264/Advanced Video Coding

Scalar quantizer of luma and chroma blocks

- **Includes scaling**

$$Z_{ij} = \text{round}\left(W_{ij}, \frac{PF}{QStep}\right)$$

$$\mathbf{W} = \mathbf{C}_f \mathbf{X} \mathbf{C}_f^T$$

Position	*PF*
(0,0), (2,0), (0,2) or (2,2)	a^2
(1,1), (1,3), (3,1) or (3,3)	$b^2/4$
Other	$ab/2$

- **Quantization parameter QP** for **each macroblock**

QP	0	1	2	3	4	5	6	7	8	9	10	11	12	...
QStep	0.625	0.6875	0.8125	0.875	1	1.125	1.25	1.375	1.625	1.75	2	2.25	2.5	
QP	...	18	...	24	...	30	...	36	...	42	...	48	...	51
QStep		5		10		20		40		80		160		224

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H.264/Advanced Video Coding

- **(B) Transform** of **4x4 luma DC coefficients**
  - For 16x16 Intra macroblock **high correlation** among **DC coefficients** – **decorrelation** using **Hadamard transform**

$$\mathbf{Y}_D = \left(\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} [\mathbf{W}_D] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \right) / 2$$

- **(C) Transform** of **2x2 chroma DC coefficients**
  - **Decorrelation** using **Hadamard transform**

$$\mathbf{Y}_D = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} [\mathbf{W}_D] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

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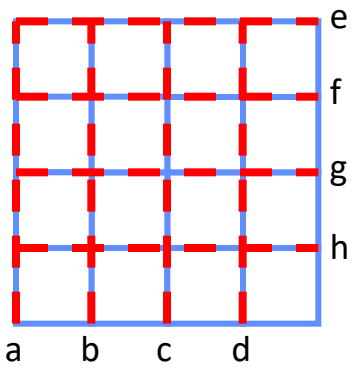
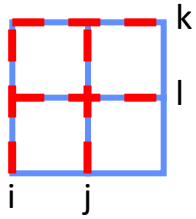
H.264/Advanced Video Coding

- **Deblocking filter – reduction** of **blocking distortion**
  - Applied **adaptively** on each **decoded macroblock**
  - Two **main effects**
    - **Smoothing** of **block edges** – **better** subjective **quality**
    - **Smaller residual error** for **motion compensated prediction**
  - **Bitrate reduction** of about **5** to **10%**



16x16 luma



8x8 chroma

Edge filtering order:
(1) Filtering vertical edges
(2) Filtering horizontal edges

Maximum of **3 samples around edge affected**

H.264/Advanced Video Coding

- **Deblocking filter – reduction** of **blocking distortion**
  - Applied to **frame** with **high compression** ratio

Without filtering

With filtering

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H.264/Advanced Video Coding

- Two main **entropy encoding principles supported**
  - **VLC** (**Variable Length Coding**) using so-called **exp-Golomb codeword table**
    - For each element **excluding quantized coefficients**

code_num	Codeword
0	1
1	010
2	011
3	00100
4	00101
5	00110
6	00111
7	0001000
…	…

  - **CAVLC** (Context-Adaptive VLC) – **switching VLC tables** for variety of **elements** and **according** to **statistic description** of **already transferred ones**
    - **Quantized coefficients** – zig-zag block readout 4x4 and 2x2

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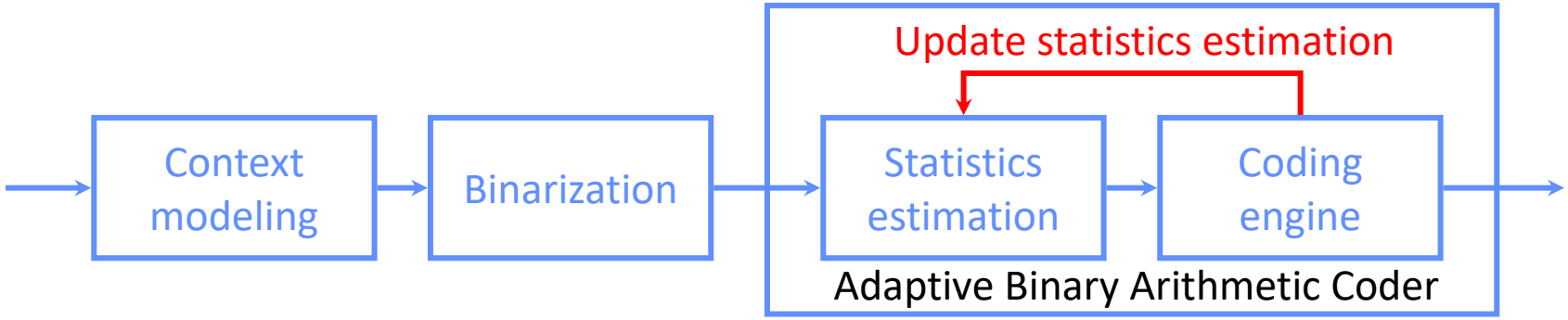
H.264/Advanced Video Coding

- **Coding efficiency** further **improved** with **CABAC**
  - **CABAC** (Context-Adaptive Binary Arithmetic Coding)
    - **Arithmetic coding** where **symbol** – **non-integer number** of **bits**
    - **Good performance** for **symbol probability >> 0.5**
    - **Adaptation** on **varying symbol statistics**
    - Bit-rate **reduction** of about **10** to **15%** compared to **CAVLC**
  - Fast **binary codec**
    - Based on **table look-ups** and **shifts** only
    - **Patented solution** (licencing fees)



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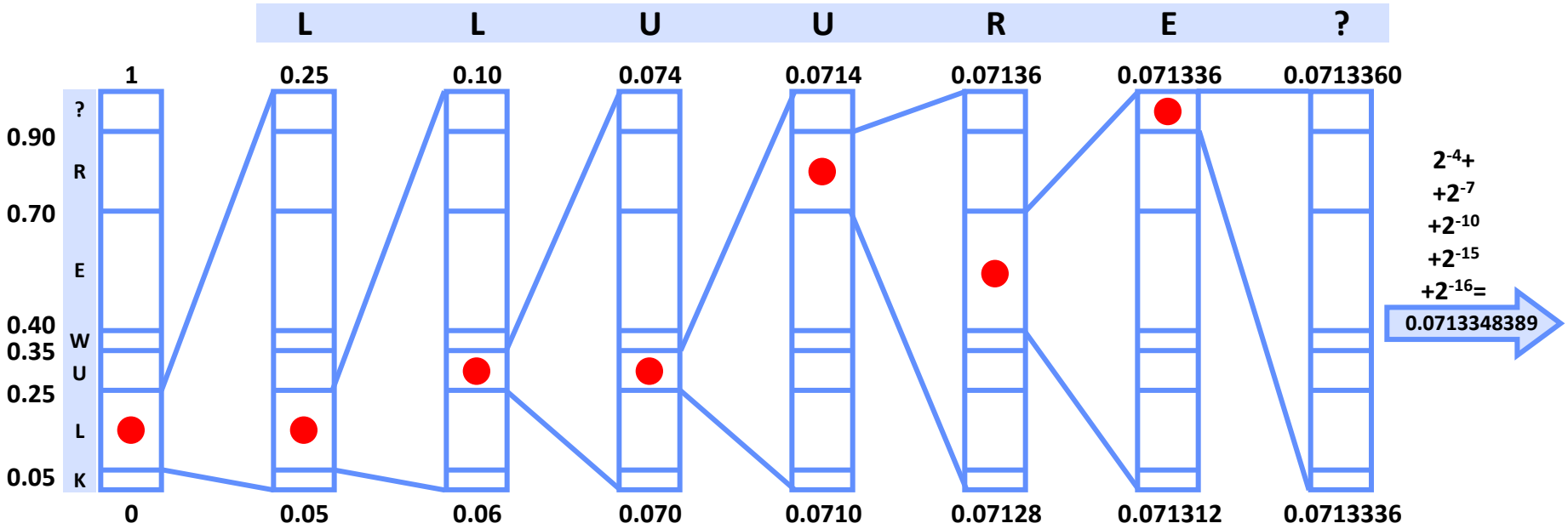
H.264/Advanced Video Coding

- **Huffman** coder
  - Input **symbol - codeword**, 1bit/symbol – **limiting efficiency**
  - **Modeling** and **coding** stages – **problematic adaptive coding**
- **Arithmetic** coder
  - Works with **sequence** of **symbols**
  - **Separates modeling** and **coding** – **dynamically adaptive**



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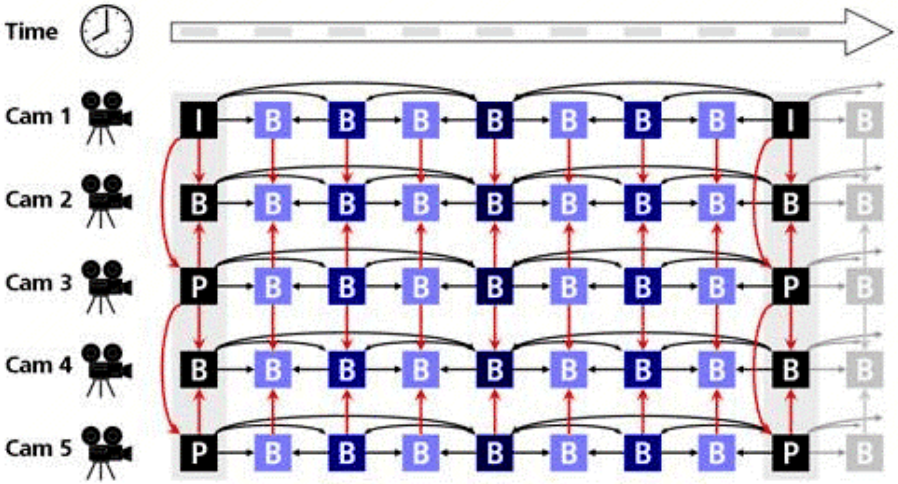
Multiview video coding

Multiview video coding (MVC)

- **Extension** of **MPEG-4 AVC standard**
  - Coding **Free Viewpoint Video** (**FVV**) – user **freely selects view**
  - **Correlation** among **frames** and **scene views** exploited
  - **MVC** for **3D movies** on **Blu-ray media**
  - **Interframe** (**time** and **interview**) **prediction** and **compensation**

- **Typical profiles**
  - **Multiview High Profile**
    - Supports **multiple views**
    - **No** support for **interlaced video**
  - **Stereo High Profile**
    - Restricted to **two views**
    - Supports **interlaced video**



Motivation for UHDTV

Performance of IT components and video technology

Year	Capacity HDD	Speed CPU	Network bandwidth	Year	Resolution	Year	Interface
	~25000x in 25 years	**~3800x** in 25 years	**~4000x** in 25 years		**~20x** in 25 years		**~45x** in 25 years
1990	340MB	I486DX 0.03 GFLOPS	10 MBps	1960	PAL/SD (720x576)	1990	SDI (270 Mbps)
1995	9GB	Pentium 200 0.2 GFLOPS	100 MBps			1995	SDI (270 Mbps)
2000	40GB	Pentium III 2.33 GFLOPS	1 GBps	2000	HD (1920x1080)	2000	HD SDI (1.5 Gbps)
2005	250GB	Pentium IV 5.6 GFLOPS	1 GBps			2002	Dual link HD SDI (2.97 Gbps)
2010	2TB	Core i7-950 49 GFLOPS	10 GBps	2010	UHD (3840x2160)	2006	3G SDI (2.97 Gbps)
2015	8TB	Core i7-6700K 114 GFLOPS	40 GBps			2015	5G SDI (6 Gbps) 12G SDI (12Gbps - 2160p60)

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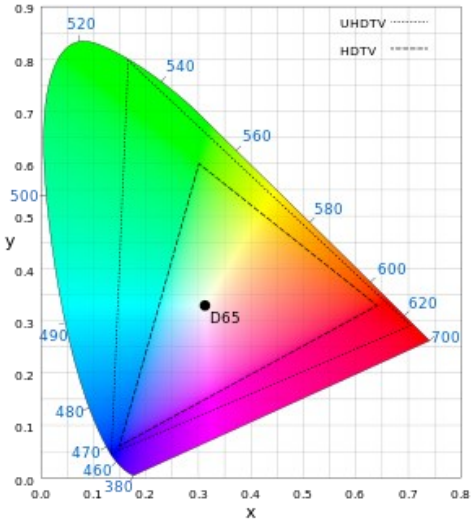
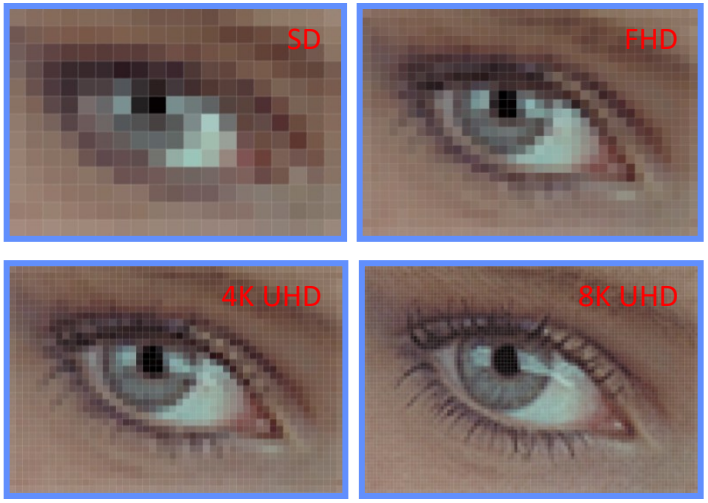
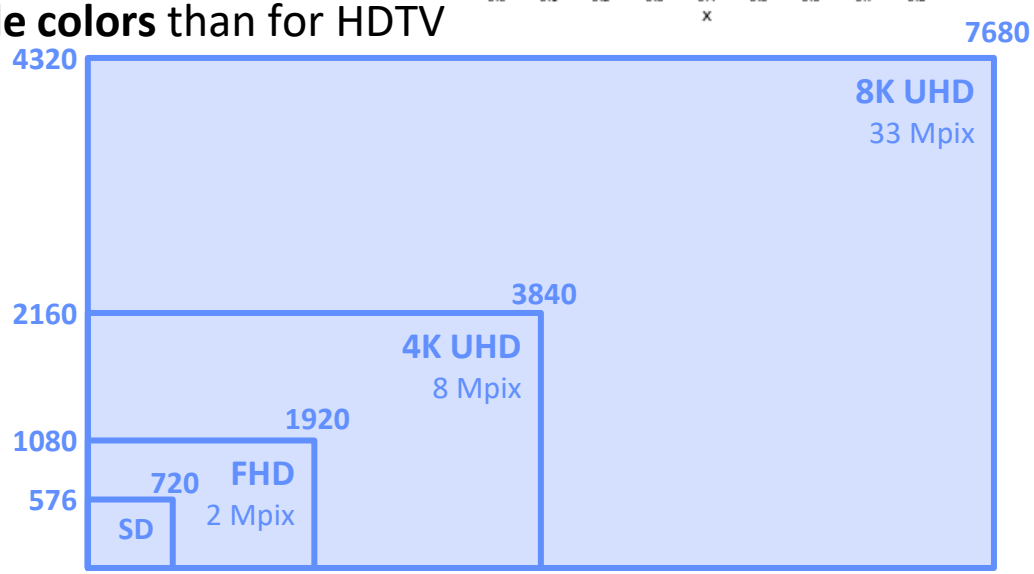
Motivation for UHDTV

Ultra high definition systems (UHDTV)

- Requirement for **higher framerate**
  - **Lowering viewing distance**
    - 1.5 H for 4K and 0.75 H for 8K
    - **Risk** of **jerky motion perception**
- Possibility of **wide color gamut** (**WCG**)
  - **Wide gamut** for **UHDTV (ITU-R BT.2020)**
    - **Broader range** of **reproducible colors** than for HDTV







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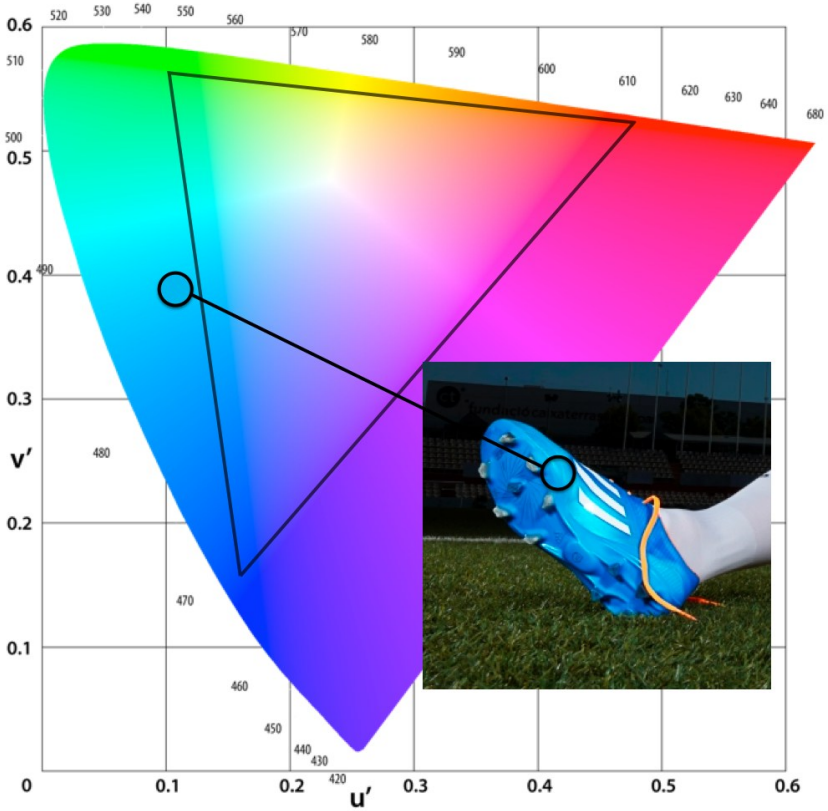
Motivation for UHDTV

- **Wide color gamut (WCG)** systems
  - Lionel Messi his **blue boots** (Adidas Samba designed for FIFA World Cup 2014)
    - **Highly saturated blue boot out** of **color range** for **HDTV system** according to **ITU-R Rec. BT.709**



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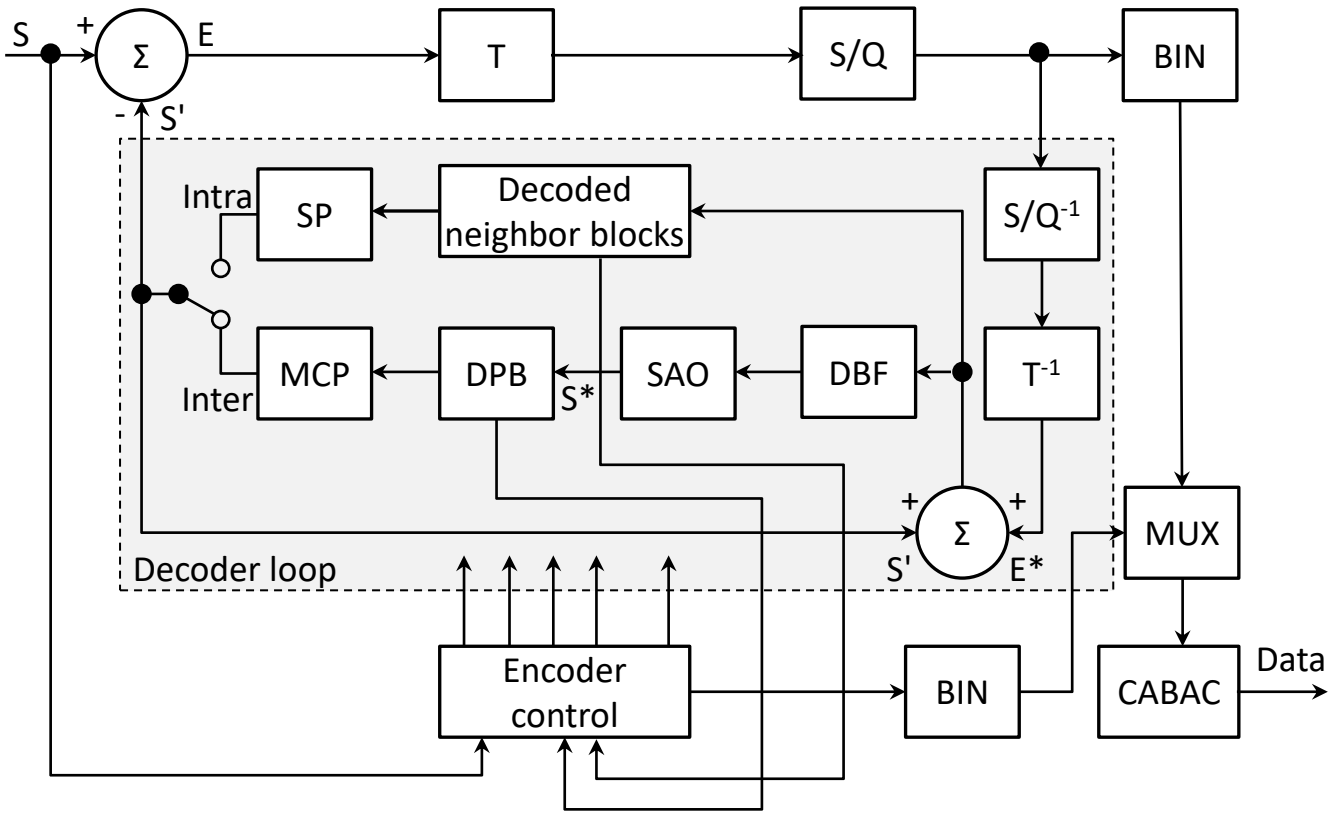
H.265/High Efficiency Video Coding

- **Motivation** for more **efficient video compression**
  - Continuously **increasing video resolution**
    - Transition **from HD to UHD**
    - In **mobile** devices **increasing HD applications**
    - Transmission of **stereoscopic** and **multiview** video
  - **Availability** of **UHD acquisition** and **display** devices
  - Selected **areas responsible** for **bitrate growth**
    - **Spatial resolution** (4K, 8K, ...), **temporal resolution** (framerate)
    - **Wide color gamut** (**WCG**) and **high dynamic range** (**HDR**)
    - **Multiview** systems
  - Required **bitrate increase faster** than **network bandwidth**
  - Necessary **development** of **efficient coding techniques**

H.265/High Efficiency Video Coding

Block diagram of hybrid HEVC encoder



S, S', S* - original picture, prediction, reconstruction
E, E* - prediction error picture (PEP), reconstructed (PEP)
T - 2D transform with variable block size
S/Q - scaling/quantization
DPB - decoded picture buffer
MCP - motion compensated prediction
SP - spatial intrapicture prediction
DBF - deblocking filter

SAO - sample adaptive offset filter
BIN - binarization
MUX - multiplexer
CABAC - context adaptive binary arithmetic coding

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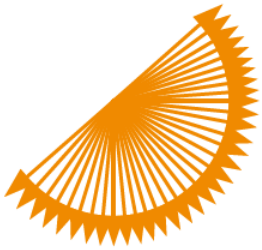
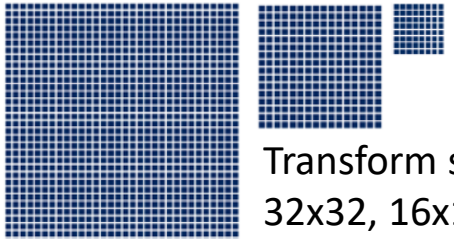
H.265/High Efficiency Video Coding

High level tool comparison AVC and HEVC

AVC	HEVC
Macroblock size 16x16	Block size 64x64
Various interframe prediction partitioning down to 4x4	Hierarchical quad-tree partitioning down to 8x8 (4x4 transform units)
9 intraframe prediction modes	 35 intraframe prediction modes
Transform sizes 8x8 and 4x4	 DST Transform sizes 32x32, 16x16, 8x8 and 4x4

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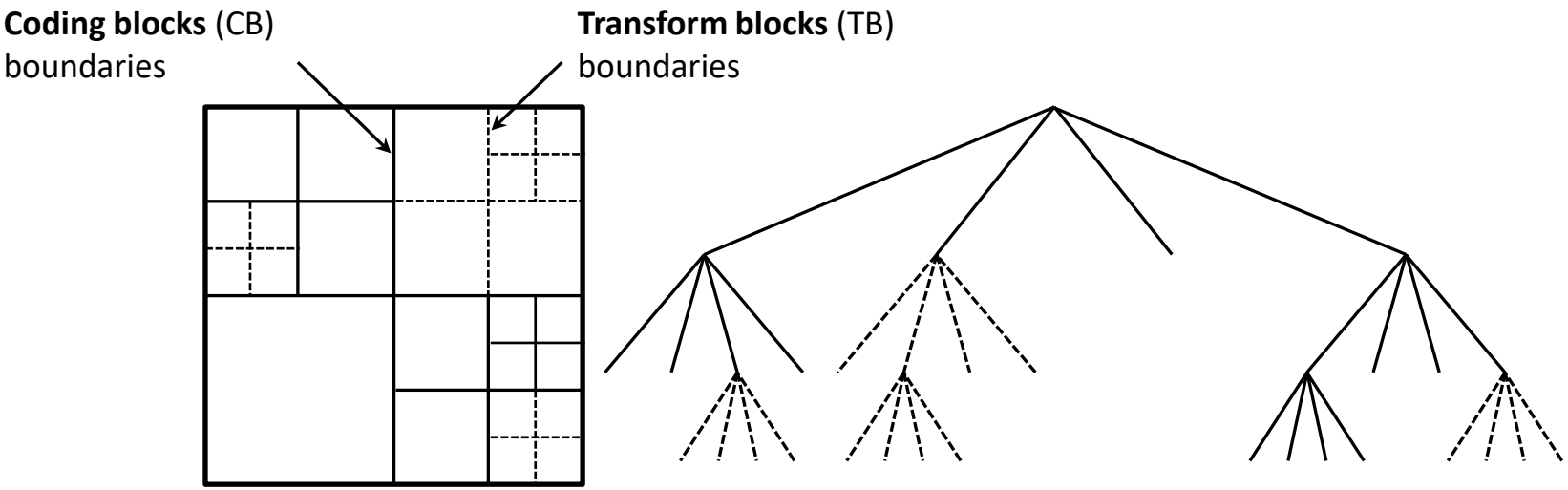
H.265/High Efficiency Video Coding

Block structure

- CTU – **Coding Tree Unit** – picture **partitioning**
  - CTU partitioned **into square CU** (Coding Units, 8x8...64x64)
- **Below CU level prediction** (**PU**) and **transform units** (**TU**)
  - **Square/rectangular prediction** (**PB**) and **square transform** (**TB**) blocks
  - **Variable sizes** PB (4x4...64x64) and TB (4x4...32x32)

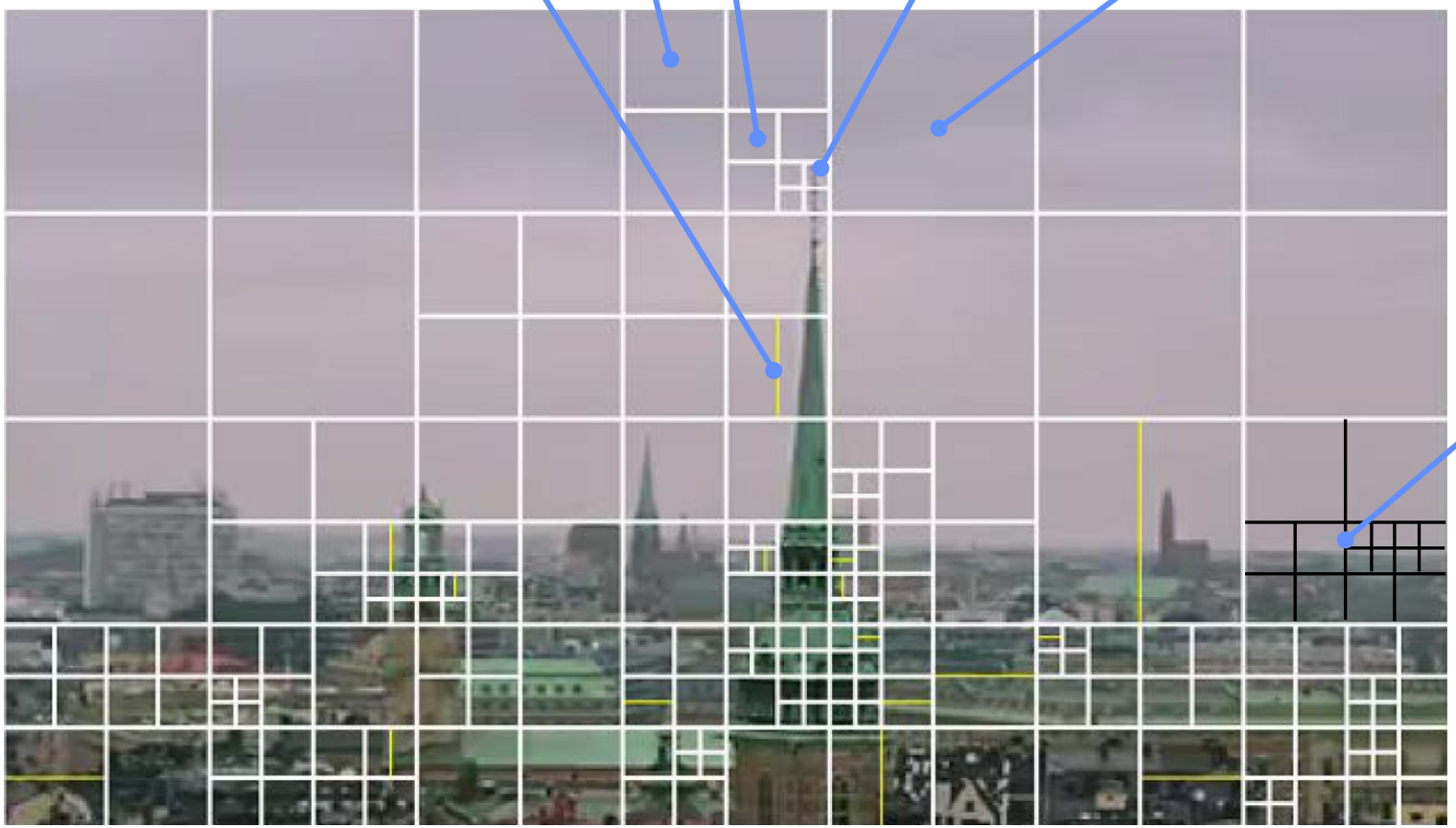


Subdivision of **coding tree block** (CTB) into **coding blocks** (CB)

Corresponding **coding tree block** (CTB) **quadtree**

H.265/High Efficiency Video Coding

Coding tree



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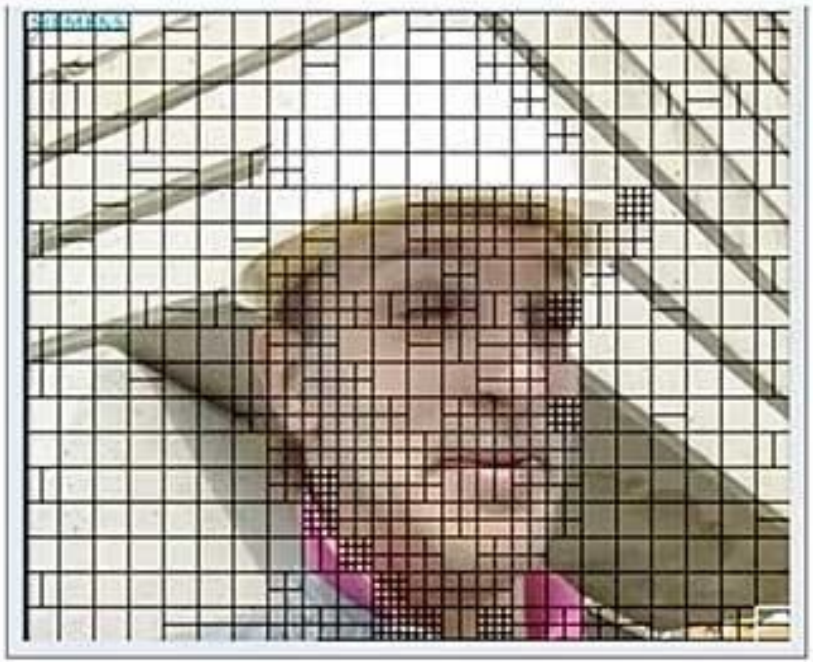
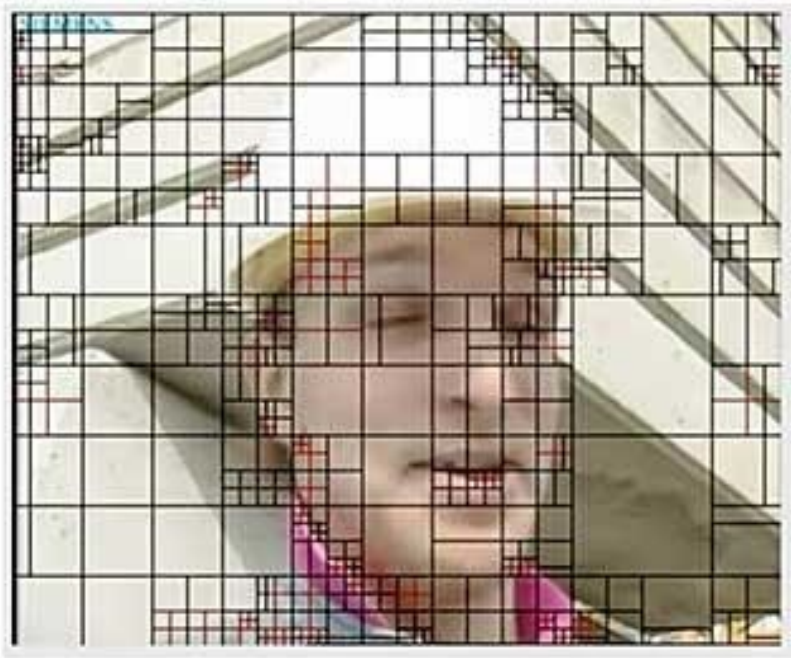
H.265/High Efficiency Video Coding

- **Partitioning** in **H.264/AVC** and **H.265/HEVC**



H.264/AVC

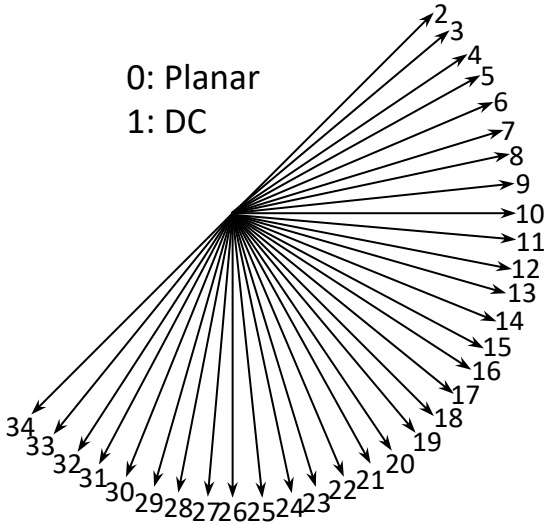


H.265/HEVC

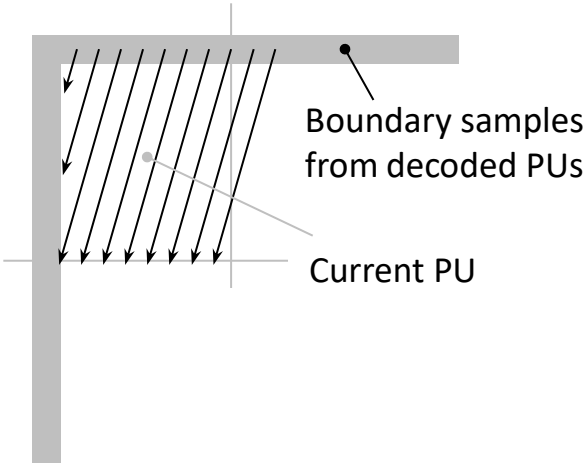
H.265/High Efficiency Video Coding

Intrapicture prediction

- Directional **prediction** from **already decoded samples**
  - Directional prediction for **33 different angles**
  - **DC** and **planar more** also available



Modes and directional orientations for intrapicture prediction



Example for directional mode 29

  - **Prediction mode** has to be **signalized** to **decoder**
    - **Mode signalized** for **each prediction unit** (**PU**)
    - **Prediction** performed **independently** for **given PU**

H.265/High Efficiency Video Coding

- **Profiles** and **levels** in **HEVC standard**
  - Similarly as for **previous standards**
    - **Profile** defines **set** of **coding tools**
    - **Level** defines e.g. **resolution**, **bitrate**, …
  - **Bitrate** for given combination **profile/level**
    - Must be **decodable** using any **decoder** according to **profile/level**
  - **Three profiles defined** in **first version**
    - **Main** – **common consumer video** devices (support YCbCr 4:2:0, 8bps)
    - **Main 10** – **higher bit depths** (up to 10 bits per sample)
    - **Main Still Picture** – **still picture** encoding (intrapicture coding only)
  - **Extension profiles** in **second version** (21 profiles)
    - Monochrome, Monochrome 12, Monochrome 16, Main 12, Main 4:2:2 10, Main 4:2:2 12, Main 4:4:4, Main 4:4:4 10, Main 4:4:4 12, Main Intra, Main 10 Intra, Main 12 Intra, Main 4:2:2 10 Intra, Main 4:2:2 12 Intra, Main 4:4:4 Intra, Main 4:4:4 10 Intra, Main 4:4:4 12 Intra, Main 4:4:4 16 Intra, Main 4:4:4 Still Picture, Main 4:4:4 16 Still Picture

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H.265/High Efficiency Video Coding

Profiles and levels in HEVC standard

- Further **division** into **tiers** and **levels**
  - Defining **13 levels** denoted 1, 2, 2.1, 3, 3.1, 4, 4.1, 5, 5.1, 5.2, 6, 6.2
  - For **level 4** and higher **two tiers** defined
    - **Main** – lower tier for **most applications**
    - **High** – higher tier for **very demanding applications**
- **Enormous range** of **video parameters**
  - **Range** of **picture resolutions**
    - **SQCIF** (**128 x 96**), QCIF (176 x 144), QVGA (320 x 240), 525 SIF (352 x 240), CIF (352 x 288), 525 HHR (352 x 480), 625 HHR (352 x 576), Q720p (640 x 360), VGA (640 x 480), 525 4SIF (704 x 480), 525 SD (720 x 480), 4CIF (704 x 480), 625 SD (720 x 576), 480p (16:9) (864 x 480), SVGA (800 x 600), QHD (960 x 540), XGA (1024 x 768), 720p HD (1280 x 720), 4VGA (1280 x 960), SXGA (1280 x 1024), 525 16SIF (1408 x 1152), 16CIF (1408 x 1152), 4SVGA (1600 x 1200), 1080 HD, 2Kx1K (2048 x 1024), 2Kx1080 (2048 x 1080), 4XGA (2048 x 1536), 16VGA (2560 x 1920), 3616 x 1536 (2,35:1), 3672 x 1536 (2,39:1), 3840 x 2160 (4*HD), 4Kx2K (4096 x 2048), 4096 x 2160, 4096 x 2304 (16:9), 4096 x 3072, 7680 x 4320, 8192 x 4096, **8192 x 4320**
  - **Level defines** also **other parameters**, e.g., framerate

H.265/High Efficiency Video Coding

Level	Maximum luma sample rate [samples/s]	Maximum luma picture size [samples]	Maximum bitrate for Main and Main 10 profile [kbit/s]		Maximum bitrate for Main 12 profile [kbit/s]		Maximum bitrate for Main 4:4:4 12 profile [kbit/s]		Maximum bitrate for Main 4:4:4 16 Intra profile [kbit/s]		Maximum bitrate for High Throughput 4:4:4 16 Intra profile [kbit/s]		Example picture resolution at highest frame rate
			Main tier	High tier	Main tier	High tier	Main tier	High tier	Main tier	High tier	Main tier	High tier	
1	552,960	36,864	128	-	192	-	384	-	1,024	-	12,288	-	176×144 @15.0
2	3,686,400	122,880	1,500	-	2,250	-	4,500	-	12,000	-	144,000	-	352×288 @30.0
2.1	7,372,800	245,760	3,000	-	4,500	-	9,000	-	24,000	-	288,000	-	640×360 @30.0
3	16,588,800	552,960	6,000	-	9,000	-	18,000	-	48,000	-	576,000	-	960×540 @30.0
3.1	33,177,600	983,040	10,000	-	15,000	-	30,000	-	80,000	-	960,000	-	1280×720 @33.7
4	66,846,720	2,228,224	12,000	30,000	18,000	45,000	36,000	90,000	96,000	240,000	1,152,000	2,880,000	2,048×1,080 @30.0
4.1	133,693,440		20,000	50,000	30,000	75,000	60,000	150,000	160,000	400,000	1,920,000	4,800,000	2,048×1,080 @60.0
5	267,386,880	8,912,896	25,000	100,000	37,500	150,000	75,000	300,000	200,000	800,000	2,400,000	9,600,000	4,096×2,160 @30.0
5.1	534,773,760		40,000	160,000	60,000	240,000	120,000	480,000	320,000	1,280,000	3,840,000	15,360,000	4,096×2,160 @60.0
5.2	1,069,547,520		60,000	240,000	90,000	360,000	180,000	720,000	480,000	1,920,000	5,760,000	23,040,000	4,096×2,160 @120.0
6	1,069,547,520	35,651,584	60,000	240,000	90,000	360,000	180,000	720,000	480,000	1,920,000	5,760,000	23,040,000	8,192×4,320 @30.0
6.1	2,139,095,040		120,000	480,000	180,000	720,000	360,000	1,440,000	960,000	3,840,000	11,520,000	46,080,000	8,192×4,320 @60.0
6.2	4,278,190,080		240,000	800,000	360,000	1,200,000	720,000	2,400,000	1,920,000	6,400,000	23,040,000	76,800,000	8,192×4,320 @120.0

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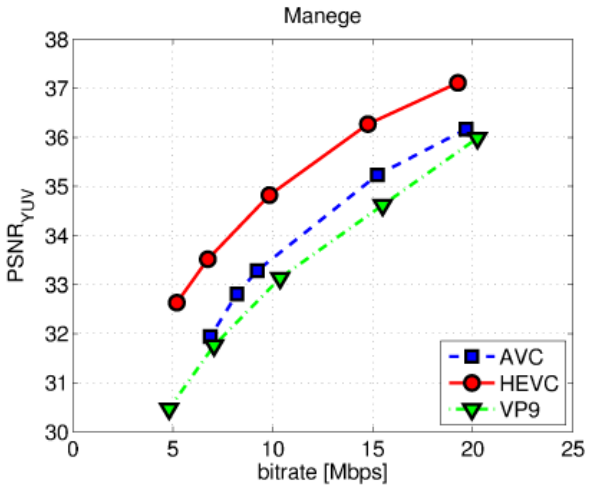
H.265/High Efficiency Video Coding

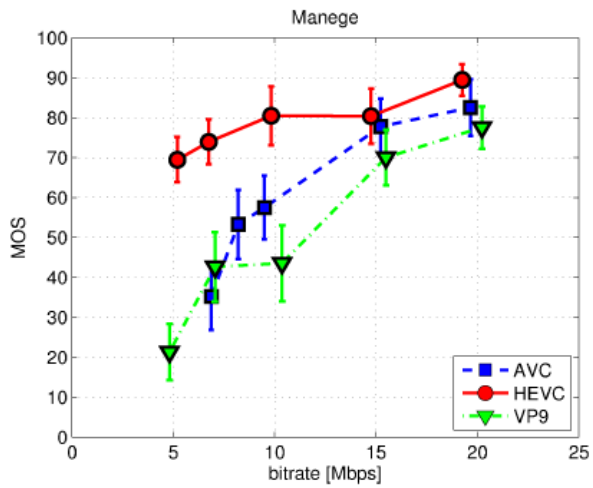
- **Other technologies** besides **ITU-T** and **ISO/IEC**
  - **Google** developing **VP8**, **VP9** codecs
    - Primarily **intended** for **web applications**
    - Video **compression** using **VP8** and **VP9**
    - **VP8** initially developed by **On2 Technologies** → **Google** (2010)
    - **VP9** targeting **50% bitrate savings** with **respect** to **VP8**
    - Available **studies** show **VP9 performs worse** than **HEVC**
  - Foundation **Xiph.Org** (supported **Mozilla**) develops **Daala**
    - Targeting **open-source**, **royalty-free format**
    - **Better performance** than **HEVC** and **VP9**
    - Exploits **transform** with **overlapping blocks**
    - Substantial **blocking artifact suppression**
    - Optimized **vector quantization**

H.265/High Efficiency Video Coding

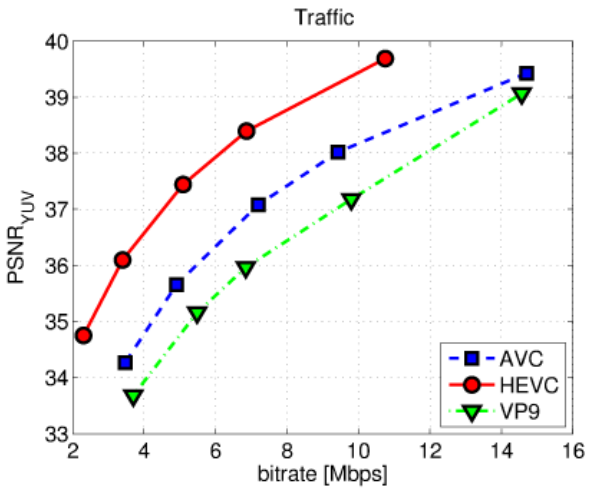
Comparing compression performance

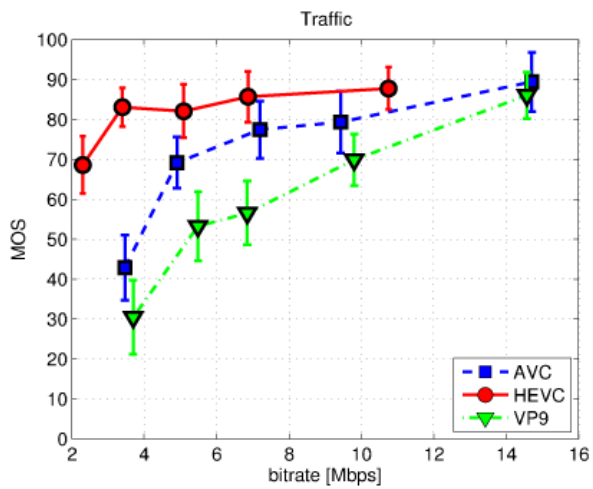
- **UHD applications** (subjective and objective assessment)

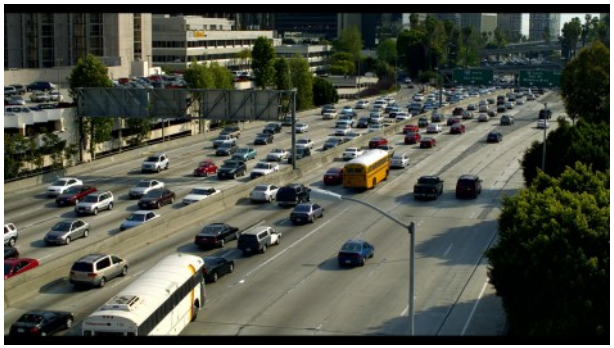




4K test sequence







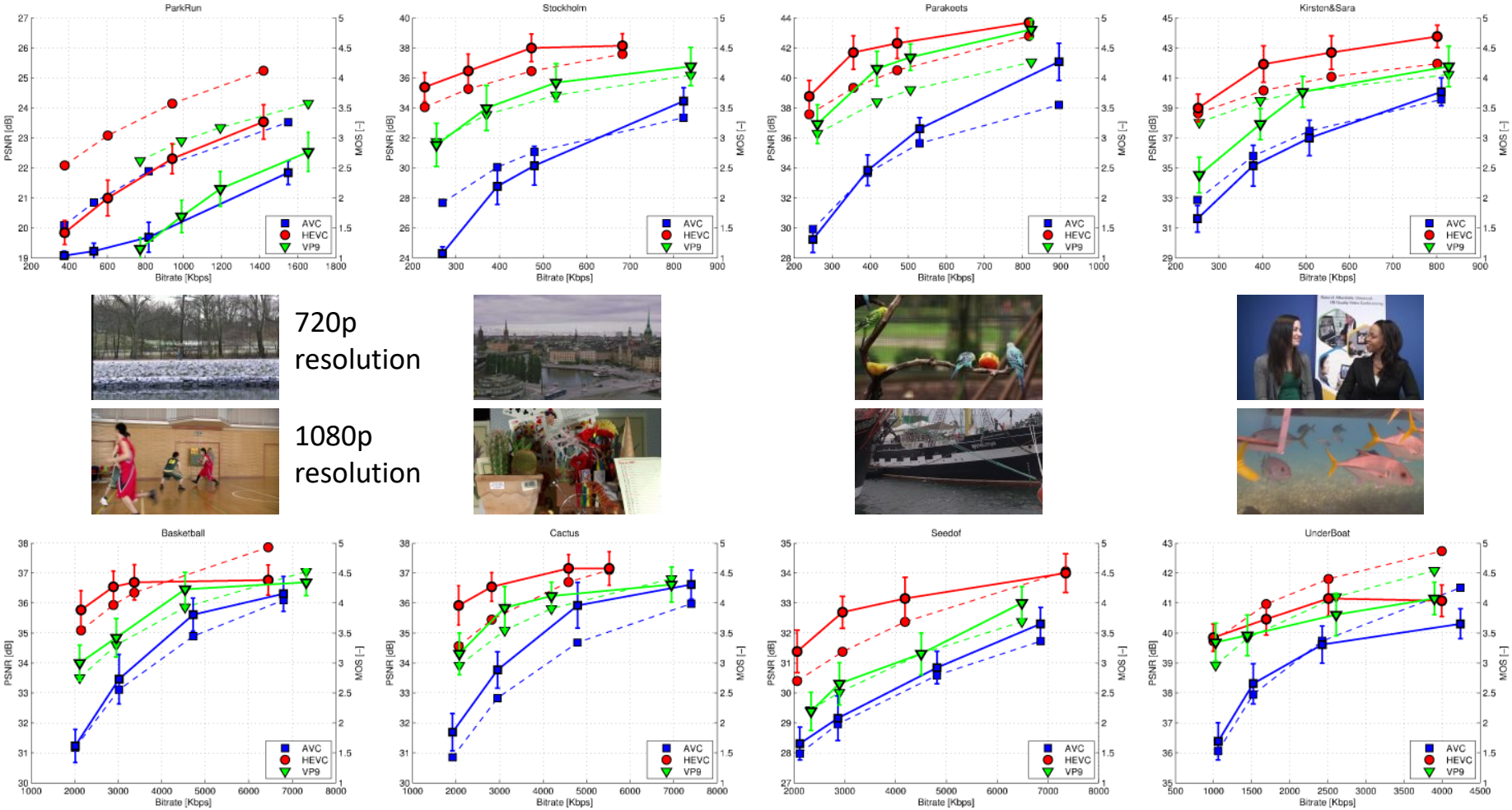
4K test sequence

Řeřábek, M., Ebrahimi, T. Comparison of compression efficiency between HEVC/H.265 and VP9 based on subjective assessments, Proceedings of SPIE 9217, 2014.

H.265/High Efficiency Video Coding

Comparing compression performance

- **Real-time applications** (subjective and objective assessment)



Řeřábek, M., Hanhart, P., Korshunov, P., Ebrahimi, T. Quality evaluation of HEVC and VP9 video compression in real-time applications, QoMEX 2015.

H.265/High Efficiency Video Coding

Comparing compression performance

- **UHD applications** (subjective and objective assessment)

Content	HEVC vs AVC		VP9 vs AVC		HEVC vs VP9	
	BD-PSNR	BD-MOS	BD-PSNR	BD-MOS	BD-PSNR	BD-MOS
Manege	-28.7%	-44.6%	10.6%	29.2%	-39.7%	-63.7%
Traffic	-37.2%	-57.5%	25.1%	61.0%	-49.8%	-74.7%
Tree Shade	-22.7%	-37.4%	18.9%	-8.2%	-33.7%	-31.9%
Sintel2	-69.9%	-70.9%	-60.9%	-61.7%	-19.0%	-27.5%
Average	**-39.6%**	**-52.6%**	**-1.59%**	**5.1%**	**-35.6%**	**-49.4%**

- **Real-time applications** (subjective and objective assessment)

Content	HEVC vs AVC		HEVC vs VP9		VP9 vs AVC	
	BD-Rate	ΔR	BD-Rate	ΔR	BD-Rate	ΔR
ParkRun	-54.8 %	-53.0 %	-44.0 %	-46.2 %	-18.3 %	-7.4 %
Stockholm	-	-	-46.1%	-54.7%	-55.9 %	-49.7 %
Parakeets	-69.1 %	-62.4 %	-32.1 %	-28.0 %	-55.5 %	-48.5 %
Kirsten&Sara	-60.1 %	-62.6 %	-20.8 %	-43.8 %	-52.8 %	-30.8 %
Basketball	-55.8 %	-59.3 %	-38.3 %	-45.6 %	-31.5 %	-28.3 %
Cactus	-54.3 %	-57.5 %	-23.6 %	-43.0 %	-42.9 %	-31.9 %
Seedof	-52.8 %	-61.8 %	-36.0 %	-51.1 %	-26.9 %	-21.9 %
Underboat	-54.2 %	-60.1 %	-27.6 %	-26.6 %	-39.2 %	-48.1 %
Average	**−57.3%**	**−59.5%**	**−33.6%**	**−42.4%**	**−40.4%**	**−33.3%**

Řeřábek, M., Ebrahimi, T. Comparison of compression efficiency between HEVC/H.265 and VP9 based on subjective assessments, Proceedings of SPIE 9217, 2014.
Řeřábek, M., Hanhart, P., Korshunov, P., Ebrahimi, T. Quality evaluation of HEVC and VP9 video compression in real-time applications, QoMEX 2015.

H.265/High Efficiency Video Coding

- **Overview** of **main HEVC features**
  - **Reducing bitrate** to **half** with **respect** to **H.264/AVC**
    - Substantial **computational complexity increase**
  - Intended for **recent acquisition** and **display devices**
    - Picture resolution range **320 x 240** (QVGA) to **8192 x 4320** (8K UHD)
    - Progressive sampling, wide **gamut** and **high dynamic range**
  - Exploits **hybrid encoder**
    - **Intraframe** and **interframe** prediction
      - 2D **transform coding**
    - **Macroblocks substituted** with **coding tree units** (**CTU**)
      - Allows **picture partitioning** to **variable areas**
    - **Entropy encoding** based on **CABAC only**
    - More **accurate interframe prediction**
    - Improved **deblocking filter** and **sample adaptive offset filter**
  - Support for **parallel processing** and **efficient computing**

H.265/High Efficiency Video Coding

Extensions to HEVC

Including extensions for special applications

- **3D – 3D video compression** (finalized in **2015** as **3D-HEVC**)
  - Efficient **depth map representation** (exploiting redundancy between the depth maps and picture)
  - **Motion vectors** utilized also for **depth maps**
  - **Depth map** exploited for more accurate **interview prediction**
- **HDR** (High Dynamic Range) – high dynamic range video
  - HDR for better **quality of experience** (**MPEG** tackles since **2015**)
    - **Standard dynamic range** (**SDR**) – up to **10 f-stops**
    - **Extended dynamic range** (**EDR**) – up to **10** to **16 f-stops**
    - **High dynamic range** (**HDR**) – up to **16 f-stops**
- **WCG** (Wide Color Gamut) – wide color gamut video
  - Support for **color gamuts** beyond ITU-R BT.601 (SDTV) and BT.709 (HDTV)
  - Considered **WCG** namely **ITU-R BT.2020** or **P3D65** (digital cinema)
- Important issue of **backward compatibility**

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H.266/Versatile Video Coding

- **H.266/Versatile video coding (VVC)**
  - **ITU-T/ISO** Joint Video Experts Team (**JVET**)
    - **Joint project** of ITU-T (VCEG) and ISO/IEC (MPEG)
    - First version finalized in **July 2020**
- **Main requirements** defined for **VVC project**
  - **Coding efficiency**
    - **Improvement** of **50%** over **H.265/HEVC**
    - **Resolutions** from **HD** to **UHD-I** (~4K) and **UHD-II** (~8K)
    - Video compression of **high dynamic range** HDR / 10bit
  - **Versatility**
    - Typical **screen content**
    - **Adaptive resolution change**
    - **Independent sub-pictures**

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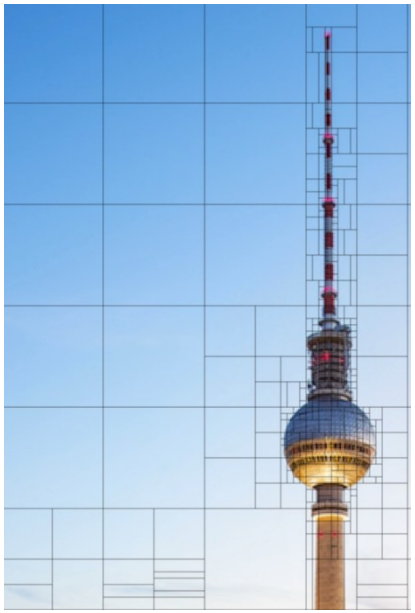
H.266/Versatile Video Coding

VVC – Development

- **Draft 1** First Test Model (VTM-1.0)
  - **Selected technologies** for further development
    - **Including block partitioning** based on **quadtree** (**QT**) plus other **multi-type tree** (**MTT**)
    - **Fundamental impact** on all **coding tools** (CfP)
  - **Test promising coding tools** from **proposals** (CfP)
    - **Adding** tools for **bitrate reduction** (**efficiency/complexity**)
- **Draft 6** (VTM-6.1)
  - **Many incremental improvements** of **hybrid design**
  - **New tools** for **improved coding efficiency**



Multi-type tree partitioning example

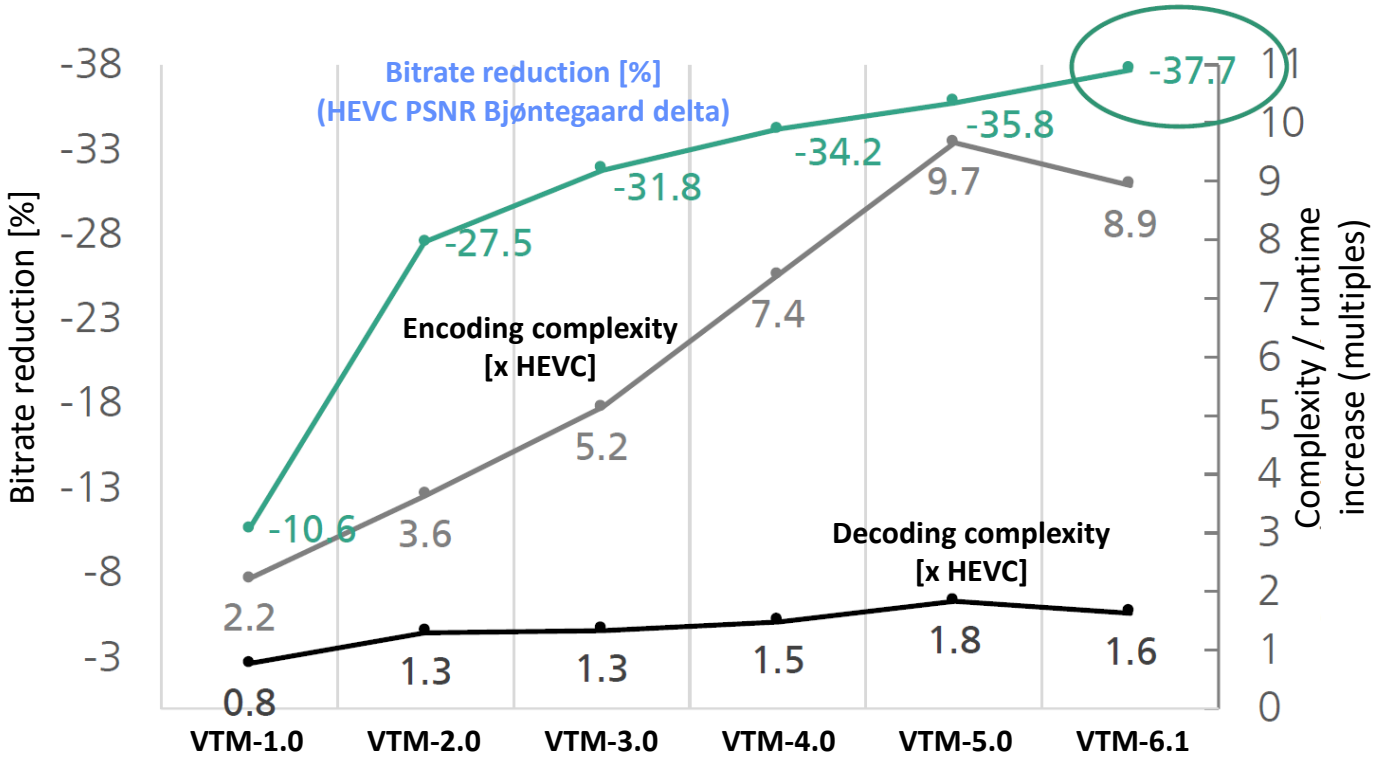
- Flexible Block Partitioning with Multi-type Tree (MTT)
- Separate Tree for Luma and Chroma (CST)
- Dependent Quantization (DQ)
- Joint coding of chrominance residuals (JCCR)
- Multiple Transform Set (MTS)
- Low frequency non-separable transform (LFNST)
- Adaptive Loop Filter (ALF)
- Affine Motion Compensation
- Subblock-based Temporal Merging Candidates
- Adaptive motion vector resolution (AMVR)
- Triangular partition mode (TPM)
- Bi-directional optical flow (BDOF)
- Merge with MVD (MMVD)
- Bi-prediction with CU weights (BCW)
- Decoder-side motion vector refinement (DMVR)
- Symmetric motion vector difference (SMVD)
- Sub-block transform (SBT)
- Combined intra/inter prediction (CIIP)
- Multi-reference line intra prediction (MRL)
- Intra block copy mode (IBC)
- Intra sub-partitioning (ISP)
- Matrix based intra prediction (MIP)
- Cross-component Linear Model (CCLM)
- Luma mapping with chroma scaling (LMCS)
- Transform Skip Residual Coding (TSRC)
- Quantized residual DPCM

Bross, B. Verstatile Video Coding (VVC) on the final stretch, 2019.

H.266/Versatile Video Coding

VVC – Coding efficiency

- **Comparing** for **reference SW implementation**
  - **VVC** (SW **VTM** – VVC Test Model) **vs. HEVC** (SW **HM** – HEVC Test Model)
  - **Similar results** as while developing **HEVC vs. AVC**
  - **Higher gain** expected in **subjective assessment**



Bross, B. Verstatile Video Coding (VVC) on the final stretch, 2019.

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VP8 and VP9

- **VP8** and **VP9** – video **compression formats** by **Google**
  - **VP8 open-source**, **royalty-free**, **successor** of **VP7**
    - Introduced by company **On2 Technologies** (**2008**)
    - Only **progressive** video, **4:2:0** color subsampling, **8bps**
    - **Traditional compression scheme**, transform coding (similar to H.264)
    - Main differences in **decoding loop** and so-called **golden frames** – **freely** selected **reference frame** from past
    - **DCT** on **blocks 4x4** followed by **WHT**, **two deblocking filters**
    - Macroblocks for **interframe prediction** 4x4, 8x8 and 16x16
  - **VP9 open-source**, **royalty-free**, **successor** of **VP8**
    - Introduced by **Google** (**2013**), exploited on **YouTube**
    - Primarily for **resolution from 1080p**, **also lossless**, support for **HDR**, ...
    - **Compression efficiency comparable** (bit worse) with **H.265/HEVC**
    - Coding **units 64x64**, motion vector accuracy 1/8, three interpolation filters, DCT 16x16 and 32x32, **DST**, improved zig-zag, ...

AOMedia Video 1 (AV1)

AOMedia Video 1 (AV1) codec

- „**Internet**“ applications, **open-source**, **royalty-free**, ...
  - **Alliance for Open Media** (AOMedia) **from 2015**
    – Amazon, Cisco, Google, Intel, Microsoft, Mozilla, Netflix, ...
  - Developed from **VP9 codec**
  - Targeting **20 % better compression** than **H.265/HEVC** and **VP9**
  - At least **50 % better compression** than **H.264/AVC**
- **Specification 1.0.0 released** in **2019**
  - **Conventional scheme** – block-based transform coding and **inter/intraframe prediction** and **compensation**
  - **Variable partitioning** in blocks from **128 x 128** up to **4 x 4**
  - Internal processing in **12 bps**, numerous **Intra prediction modes**
  - Transform of **square** and **rectangular** blocks (**DCT** and **DST**)
  - Optimized **quantization matrix**, in-loop filter (Wiener), **film grain synthesis**, **nonbinary arithmetic coding**,...

AOMedia Video 1 (AV1)
Initial release (2018)

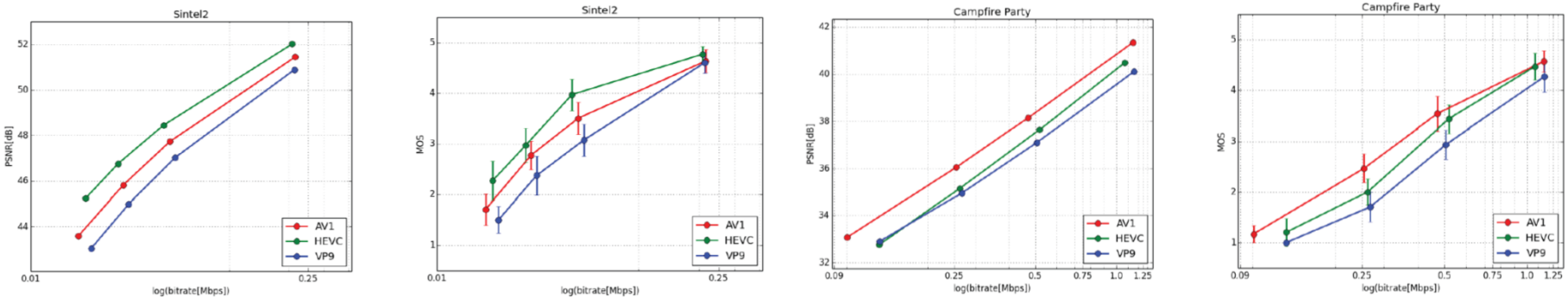
Open royalty-free, better compression performance (similar to VVC), enhanced **support** for **AR**, **VR**

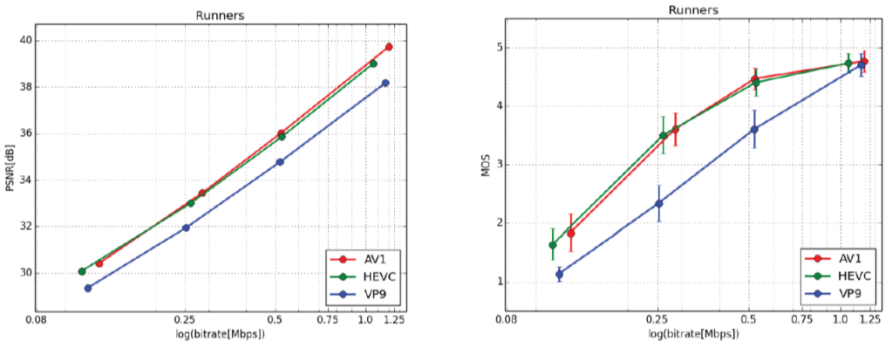
AOMedia Video 2 (AV2)
Draft status (May 2026)

AOMedia Video 1 (AV1)

AOMedia Video 1 (AV1) codec

Comparing efficiency H.265/HEVC, VP9 and AV1





Average bitrate savings for constant quality

Content	AV1 vs HEVC/H.265		AV1 vs VP9		HEVC/H.265 vs VP9	
	BD-PSNR	BD-MOS	BD-PSNR	BD-MOS	BD-PSNR	BD-MOS
Campfire Party	-23.2%	-19.0%	-30.9%	-35.4%	-9.3%	-19.6%
Runners	-2.6%	1.5%	-24.1%	-40.8%	-22.2%	-41.5%
Traffic Flow	5.7%	4.9%	-35.2%	-47.7%	-38.5%	-55.6%
Tree Shade	-5.8%	-9.3%	-26.6%	-21.8%	-22.8%	-14.7%
Sintel2	35.6%	38.0%	-23.6%	-26.7%	-43.7%	-46.7%
Average	**1.9%**	**3.2%**	**-28.1%**	**-34.5%**	**-27.3%**	**-35.6%**

Sintel2

Campfire Party

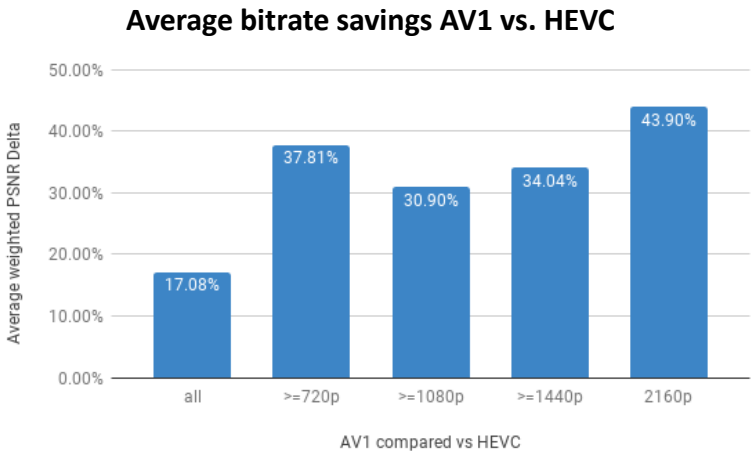
Runners

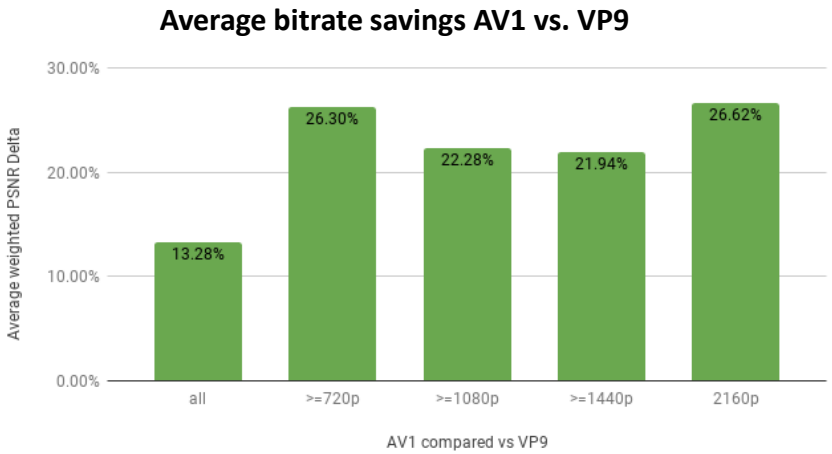
Akyazi, P., Ebrahimi, T. Comparison of Compression Efficiency between HEVC/H.265, VP9 and AV1 based on Subjective Quality Assessments, QoMEX 2018.

AOMedia Video 1 (AV1)

AOMedia Video 1 (AV1) codec

- **Comparing efficiency H.265/HEVC, VP9** and **AV1 streaming**
  - **AVC** offers **best compatibility**, **HEVC** and **AV1** better **efficiency**
  - **VP9** and **AV1** developed as **open-source**, **royalty-free**,...
  - **AVC** and **HEVC** require **royalties**, …
- **Large scale database** for **objective performance testing**
  - Video from resolution **256 x 144** up to **4K** (~**100 kbps** up to ~**20 Mbps**)
  - Exploiting **ffmpeg** implementation (libx264, libx265, libvpx-vp9)





https://bitmovin.com/av1-multi-codec-dash-dataset/

Apple ProRes

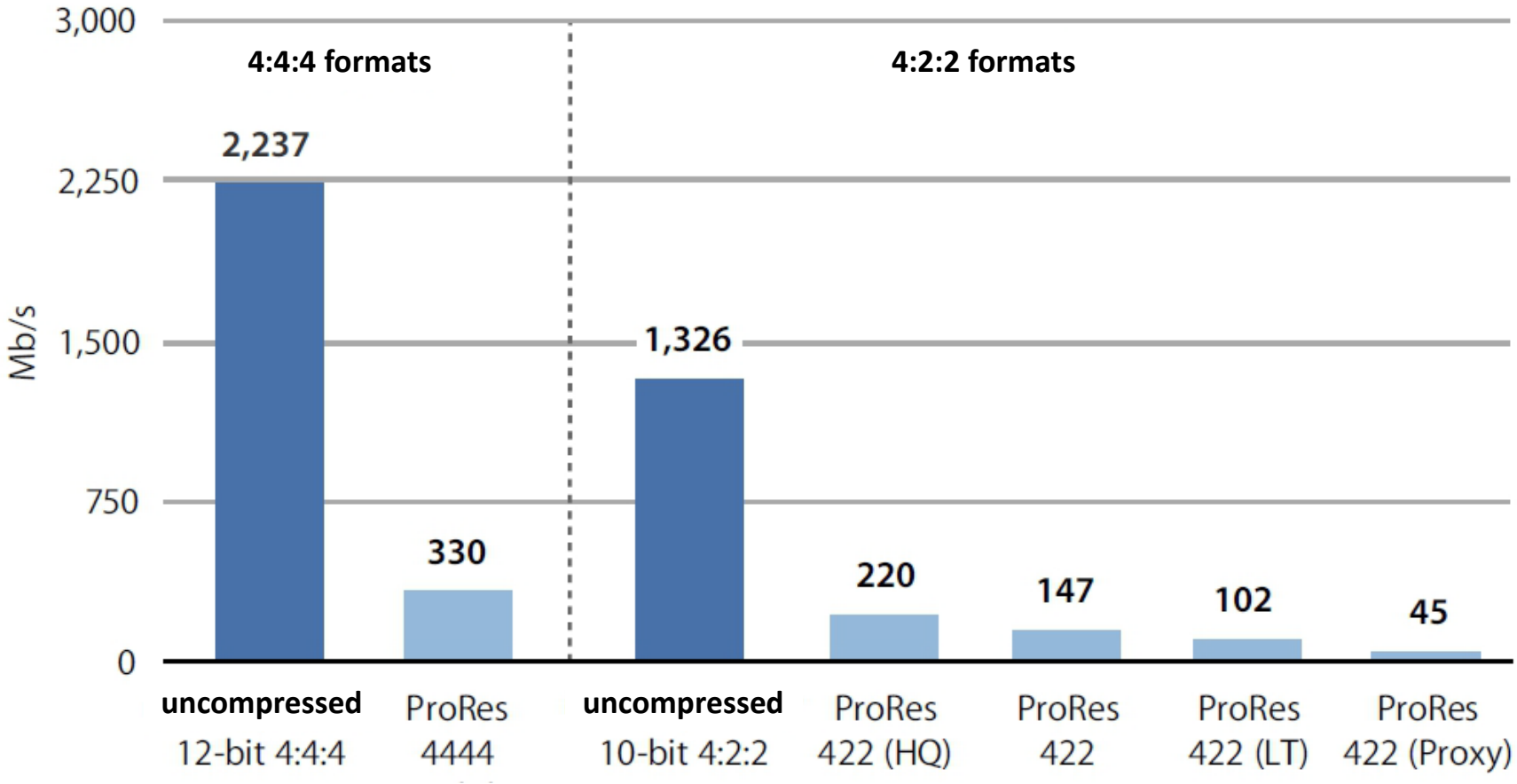
Apple ProRes – lossy video compression

- **High quality – post-production, intermediate codec**
  - Format intended **primarily** for **video editing**
  - Only **intraframe coding** exploiting **DCT block-based** approach
  - **Not intended** for **video broadcasting** and **content distribution**
  - **Large range** of **resolutions** (**720 x 576 SD** up to **8K**)
- **Major variants ProRes 422** and **ProRes 4444**
  - **ProRes 422** (variants **Proxy**, **LT**, **HQ**)
    - **Common applications**
    - **Color subsampling 4:2:2** and **10bpc**
    - Bitrate (HD) **145 Mbps** (default), **220 Mbps** (**HQ** – high quality), **100 Mbps** (**LT** – compromise), **40 Mbps** (**Proxy** – offline editing)
  - **ProRes 4444**
    - **Advanced post-production applications**, **support** for **alpha channel**
    - **Full color sampling 4:4:4** and **12bpc**, **alpha channel** (up to 16bpp)
    - Bitrate (HD) **330 Mbps**

Apple ProRes

Apple ProRes – comparing bitrates

- **Example** for **HD video** (1080p30)



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Dirac and Dirac Pro (VC-2)

- **Dirac – video compression** with **wavelet transform**
  - **Open-source**, **royalty-free**, ...
    - Developed by **BBC Research & Development** (**2008**)
  - **Dirac Pro** (**studio** applications, only **intraframe** coding)
    - **Standardized** under **SMPTE** as **VC-2** (SMPTE 2042-1:2009)
  - Intended primarily for **Ultra HDTV** applications
    - Video **resolution** up to **8K**, **framerate** from 23.97 up to **60 fps**
  - **Lossless**, **visually lossless** or **lossy compression**
  - Compression **performance comparable** with **H.264/AVC**
    - Exploiting **wavelet transform** (as JPEG 2000)
  - Two computationally **efficient reference implementations**
    - Named after theoretical physicists (Paul Dirac, Erwin Schrödinger)
    - **Dirac-research** (2008)
    - **Schrödinger** (2008)

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Video container formats

Selected formats for video files

- **Audio Video Interleave** (**AVI**)
  - **Microsoft** (1992), audio + video, limitations, ...
- **Matroska Multimedia Container** (MKV, MK3D, MKA, MKS)
  - **Open** format, audio + video + image + subtitles, broad support, ...
- **MPEG-4 Part 14** (**MP4**)
  - **Streaming** support, audio + video + image + subtitles, MPEG support, ...
- **QuickTime File Format** (**MOV**, QT)
  - **Proprietary Apple**, audio + video + obraz + titulky, GIF, MJPEG, MPEG, ...
- **WebM**
  - **Open**, audio + video, web, native HTML5, MKV, VP8, Vorbis, ...
- **DivX Media Format** (**DIVX**)
  - Also **streaming**, audio + video + image + subtitles, MPEG, ...
- **Advanced Systems Formats** (**ASF**)
  - **Windows Media Video**, Audio (WMV, WMA), proprietary MS OS Windows, ...
- **Motion-JPEG** (MJPEG), **AVCHD** and many others....

https://en.wikipedia.org/wiki/Comparison_of_video_container_formats

Recommended literature

- [1] Richardson, I. E. G.: **H.264 and MPEG-4 Video Compression**, Willey, 2003.
- [2] Bhaskaran, V., Konstantinides, K.: **Image and Video Compression Standards**, Kluwer, 1997.
- [3] Sayood, K.: **Introduction to Data Compression**, Morgan Kaufmann Publishers, 2006.
- [4] Gonzalez, R. C., Woods, R. E.: **Digital Image Processing**, Pearson, 2007.
- [5] Thyagarajan, K. S.: **Still Image and Video Compression with MATLAB**, Willey, 2011.

Thank you for your attention!

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